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United States Patent	12404307
Kind Code	B2
Date of Patent	September 02, 2025
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Methods and compositions for use of recombinant bacterial effector proteins as anti-inflammatory agents

Abstract

Provided herein are methods and compositions comprising a set of paired peptides comprising a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The paired peptides can be linked to a protein transduction domain. The compositions can be formulated as pharmaceuticals. The compositions are useful for the treatment of inflammatory disorders.

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Appl. No.: 17/686110

Filed: March 03, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220306702 A1	Sep. 29, 2022

Related U.S. Application Data

continuation parent-doc US 16964762 ABANDONED WO PCT/US2019/015224 20190125 child-doc US 17686110
us-provisional-application US 62622068 20180125

Publication Classification

Int. Cl.: C07K14/255 (20060101); A61K38/00 (20060101); C07K14/25 (20060101)

U.S. Cl.:

CPC C07K14/255 (20130101); C07K14/25 (20130101); A61K38/00 (20130101); C07K2319/03 (20130101); C07K2319/30 (20130101); C07K2319/70 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of pending U.S. patent application Ser. No. 16/964,762, filed Jul. 24, 2020, which is a national stage of International Patent Application No. PCT/US2019/015224, filed Jan. 25, 2019, which claims the benefit of the priority of U.S. Provisional Patent Application No. 62/622,068, filed Jan. 25, 2018, which applications are incorporated herein by reference.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED IN ELECTRONIC FORM

(1) Applicant hereby incorporates by reference the Sequence Listing material filed in electronic form herewith. This file is labeled "WST171USC1_SeqList.txt", created Jul. 24, 2020, and having 223 KB.

FIELD OF THE INVENTION

(2) The present invention relates to methods and compositions for treating inflammatory disorders.

BACKGROUND OF THE INVENTION

(3) Most treatments for acute inflammation, such as skin inflammatory conditions, only treat the symptoms (swelling, redness, pain, heat) by using non-specific drugs like corticosteroids and emollients/skin softeners. None of these non-specific drugs affect the underlying mechanism of activation of the inflammatory pathway, e.g., NFkB/JNK/p38. Moreover, normal aging processes chronically activate ectopically these three signal transduction pathways, resulting in inflammation. The inflammation makes these pathways a target for inhibition as part of an anti-aging/wellness program.

(4) To date, the cosmeceutical industry's approach to reducing skin inflammation has many flaws. The "Active Compounds" contained in most cremes and topicals do not have defined, specific targets in the inflamed cell/tissue the cell and are extremely complex mixtures/extracts/serums containing millions of ingredients with no defined targets, or ingredients which are not absorbed by skin and have no effect at all. Because these compositions do not target the actual pathways which are causing inflammation, they have the potential to alter many processes non-specifically leading to toxic side effects.

(5) A continuing need in the art exists for new and effective tools and methods for treating the causes of inflammation.

SUMMARY OF THE INVENTION

(6) Disclosed herein are compositions of paired peptides comprising a first bacterial effector polypeptide linked to a second bacterial effector polypeptide that are useful for treating an inflammatory disorder. Accordingly, disclosed are compositions that include a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The first and second bacterial effector polypeptides can be different, that is they can recognize a different molecular targets or modulate different inflammatory pathways. In an embodiment, the protein transduction domain and the set of paired peptides can be a fusion protein. The fusion protein can include one or more linkers. The protein transduction domain can be a YopM protein transduction domain, an SspH1 protein transduction domain, or an IpaH protein transduction domain. The first bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the second bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs.3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof can be an NLeE polypeptide or a fragment thereof. In some embodiments, the fusion protein can have an amino acid sequence forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

(7) Also provided¶ are fusion proteins comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The first and second bacterial effector polypeptides can be different, that is they can recognize a different molecular targets or modulate different inflammatory pathways. The fusion protein can include one or more linkers. The first bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group

consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the second bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof can be an NleE polypeptide or a fragment thereof. In some embodiments, the fusion protein comprising a set of paired peptides can have an amino acid sequence as set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88.

(8) Also provided are nucleic acids encoding a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. Also provided are nucleic acids encoding fusion proteins comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The nucleic acids can be contained within a vector, which can be expressed in a host cell. In one aspect, the compositions comprising a set of paired peptides can be formulated as pharmaceutical compositions.

(9) Also provided are methods of treating a subject having or at risk for an inflammatory disorder, by administering to the subject a therapeutically effective amount of a pharmaceutical composition comprising the set of paired peptides. The inflammatory disorder can be a gastrointestinal disorder including inflammatory bowel disease, Crohn's disease and the ileocolitis, ileocecal, jeunoileitis, and gastroduodenal subtypes of Crohn's disease, and ulcerative colitis. The inflammatory disorder can also be a skin disorder.

(10) Also provided are articles of manufacture, e.g., a kit. The kit can include measured amount of one or more of the compositions of the paired peptides and one or more items selected from the group consisting of packaging material, a package insert comprising instructions for use, a sterile fluid, and a sterile container.

(11) In one aspect, a composition comprises in a pharmaceutically acceptable carrier or excipient or formulation a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain, optionally linked covalently or non-covalently to a selected protein transduction domain (PTD) or penetrating peptide (CPP). In one embodiment, the composition further comprises an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP or to an additional PTD or CPP. In one embodiment, a composition comprises multiple first and additional constructs. In another embodiment, the constructs are further associated with targeting moieties directing delivery of the constructs to a selected cell or tissue.

(12) In another aspect, a recombinant polypeptide comprises a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain, optionally linked covalently or non-covalently to a selected protein transduction domain (PTD) or penetrating peptide (CPP) and an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP or to an additional PTD or CPP. The first construct is linked

covalently or non-covalently to one or more of the additional constructs in a single polypeptide. In another embodiment, the polypeptide comprises an optional linker amino acid sequence interposed between each first and additional construct. In another embodiment, the polypeptide is further associated with targeting moieties directing delivery of the polypeptide to a selected cell or tissue.

(13) In another aspect, a recombinant nucleic acid molecule is provided which encodes one of the constructs or polypeptides described herein. These nucleic acid molecules can be further associated with regulatory sequences for expressing the constructs in vivo or in vitro.

(14) In a further aspect, a pharmaceutical or cosmeceutical composition comprises as an active agent a polypeptide as described above, or a mixture of constructs as described above in a formulation suitable for delivery of the active agent into and through the layers of the skin. In one embodiment, the formulation contains a CAGE solvent (defined below) or other components suitable for topical administration.

(15) In yet another aspect, a pharmaceutical or cosmeceutical composition comprises as an active agent a polypeptide as described above, or a mixture of constructs as described above in a formulation suitable for delivery to a selected cell or tissue.

(16) In still other aspects, methods for making the compositions, constructs, polypeptides and nucleic acid molecules are provided.

(17) In yet a further aspect, a method for treating or ameliorating or suppressing an inflammatory response comprises administering to a subject in need thereof a composition, construct, polypeptide or nucleic acid molecule described herein.

(18) Still other aspects and advantages of these compositions and methods are described further in the following detailed description of the preferred embodiments thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features and advantages of the present invention will be more fully disclosed in, or rendered obvious by, the following detailed description of the preferred embodiment of the invention, which is to be considered together with the accompanying drawings wherein like numbers refer to like parts and further wherein:
- (2) FIG. 1 is a schematic of bacterial effector constructs.
- (3) FIG. 2 is a schematic of bacterial effector constructs.
- (4) FIG. 3 is a schematic of the cloning strategy for TAT-NleE WT and Mutant R107A.
- (5) FIG. 4 is a schematic of the cloning strategy for TAT-*Shigella* OSPZ.
- (6) FIG. 5 is a schematic of the cloning strategy for YopM PTD-NleE WT and Mutant R107A.
- (7) FIG. 6 is a schematic of the cloning strategy for YopM PTD-NleE VVT No Linker, PAPA Linker, GSGS Linker and for Mutant R107A.
- (8) FIG. 7 is an exemplary SDS gel showing purified fusion proteins.
- (9) FIG. 8 is a graph showing the results of a dose-response analysis of IL-6 production in cells treated with recombinant purified effector proteins.
- (10) FIG. 9 is a graph showing the results of an analysis of the effect of paired fusion proteins on TNF-alpha.
- (11) FIG. 10 is a graph showing the results of an analysis of the effect of paired fusion proteins on IL-6.
- (12) FIG. 11 is a graph showing the results of an analysis of the effect of paired fusion proteins on MCP-1.
- (13) FIG. 12 is a graph showing the results of an analysis of the effect of paired fusion proteins on IL-23.
- (14) FIG. 13 is 12 is a graph showing the results of a dose response analysis of rYopM and YopMo

on caspase 1 activity.

(15) FIG. 14 shows the uptake of FITC-TAT-NleE-WT-His protein by Hacat cells.

(16) FIG. 15 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker NleE-WT-His protein by Hacat cells.

(17) FIG. 16 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-PAPA Linker-NleE-WT-His protein by Hacat cells

(18) FIG. 17 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker-NleE-WT-His protein by Hacat cells.

(19) FIG. 18 shows the uptake of TAT-NleE fusion polypeptide into intact mouse skin.

(20) FIG. 19 shows is an two-photon microscopy image of uptake of TAT-NleE fusion polypeptide into intact mouse skin.

(21) FIG. 20 shows a two-photon microscopy image of a 10 micron slice of mouse skin showing uptake of TAT-NleE fusion polypeptide into intact mouse skin.

(22) FIG. 21 shows graphs illustrating an analysis of methylase activity of YopM PTD-YopM (L-Rich)-PAPA/GSGS-NleE-WT-His fusion proteins.

(23) FIG. 22 shows graphs illustrating an analysis of methyltransferase activity NleE-R107A mutant and YopM.

(24) FIG. 23 shows graphs illustrating a NleE and *Shigella* Methyltransferase activity assay.

(25) FIG. 24 depicts the results of an experiment comparing the effect of formulations on NleE methylation activity.

(26) FIG. 25 shows four micrograph panels showing transdermal penetration of CAGE-NleE protein.

(27) FIG. 26 is a schematic showing how inflammation develops from cell to tissue in the skin.

(28) FIG. 27 is a schematic showing the impact of a skin irritant on the NFkB pathways.

(29) FIG. 28 shows the amino acid sequence of NleE SEQ ID NO: 1.

(30) FIG. 29 shows the publicly available UniProt P17778 amino acid sequence of YopM SEQ ID NO: 2.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(31) This description of preferred embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description of this invention. The drawing figures are not necessarily to scale and certain features of the invention may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. When only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

(32) Technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application. The definitions contained in this specification are provided for clarity in describing the components and compositions herein and are not intended to limit the claimed invention.

(33) The present invention is based in part on the inventors' finding that combinations of bacterial effector polypeptides can have synergistic immunomodulatory activity. Many bacterial pathogens, including enteropathogenic *Escherichia coli* (EPEC), enterohemorrhagic *E. coli*, and *Shigella*, utilize a type III secretion system (T3SS) to deliver multiple virulence proteins directly into host cells. These virulence proteins, also referred to as effector proteins, are produced by bacteria during the infection of a eukaryotic host. The effector proteins down-regulate the host's immune system, typically at the site of the infection. Thus, effector proteins that target specific inflammatory pathways can function as immunomodulators, for example, in the treatment of inflammatory disorders. The inventors have found that combinations of effector proteins that target multiple inflammatory pathways produced augmented immunomodulatory effects.

(34) Accordingly, the invention features compositions and methods that are useful for the treatment of inflammatory disorders. The compositions can include fusion proteins comprising bacterial effector polypeptides or a fragment of a bacterial effector polypeptide. More specifically, the compositions can include a set of paired peptides configured as a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof. For ease of reading, we will not repeat the phrase "or a fragment thereof" on every occasion. It is to be understood that where we refer to a first bacterial effector polypeptide, we refer to the first bacterial like and effector polypeptide or a fragment thereof. Similarly, it is to be understood that where we refer to a second bacterial effector polypeptide, we refer to the second bacterial effector polypeptide or a fragment thereof. Also featured are compositions comprising a set of paired peptides configured as a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof, and linked to a protein transduction domain. The methods can include a method of treating a subject at risk for or having an inflammatory disorder.

(35) The compositions and methods described herein deliver a combinatorial construct of small recombinant protein effectors that directly target cellular pathways involved in inflammation, e.g., the NFkB pathway activated in skin inflammation. As described in detail below, the delivery of a recombinant construct comprising one or multiple effector proteins linked or fused to a PTD or CPP is useful to treat inflammation. Still other embodiments include the association of an optional targeting moiety, directing the construct to a specific cell or tissue type, with the effector protein and/or formulation in a topical carrier. While these compositions and methods of treatment have a number of advantages, a significant advantage is that the protein effectors used are not able to enter into the circulation. In one embodiment, as discussed in detail below, the compositions and methods involve the use of NleE incorporated into a cream or oil formulation that reduces skin inflammation and may be widely used for many applications, both as a cosmetic beauty crème to reduce redness and irritation and as a treatment of a disease or cause of skin irritation.

(36) The present compositions and methods using multiple e.g., NFkB/MAPK inhibitors, for treatment of inflammatory responses (e.g., pain, redness, swelling, heat) is based upon the fact that inflammation caused by infection, injury, auto-immunity, sunburn, aging, etc is detected at the cell membrane. Signals that are received by the skin cell are funneled thru the NFkB (sometimes MAPK) pathway. This is a relay mechanism in the cell that must be highly regulated. Shutting off the NFkB pathway is the key to controlling skin inflammation. Naturally occurring bacterial effector proteins (e.g., NleE, YopM, SSPH1) are very potent inhibitors of the NFkB system. Their sole target and purpose is to dampen the inflammatory response.

(37) As described herein, when these bacterial proteins are recombinantly engineered into a

selected polypeptide and delivered to inflamed skin via fusion with a PTD/CPP and optional targeting moiety, they abolish the inflammatory response. The combination of two or more effector proteins, each with a different substrate in the cell, may be combinatorially fused together in a single polypeptide. In some embodiments, such combination achieves a synergistic effect, which is a significant improvement in activity beyond that accomplished by delivery of a single effector.

(38) The bacterial effector, NleE, is characterized by specificity, potency and efficiency in shutting down NFkB and hence inflammatory reactions. The presence of NleE and/or a combination with other bacterial effectors in an anti-inflammatory formulation has a number of advantages. Among the advantages are extremely high substrate specificity, which results in only a very low, if any, chance of off-target effects and toxicity for therapeutic use. In one embodiment, fusing PTDs to NleE and its related effectors is useful to deliver the effector to sites of inflammation. We and others have shown that a number of recombinant effector proteins, when fused to PTDs can indeed cross the cell and tissue boundary and be taken up by cells resulting in NFkB/JNK/p38 pathway inhibition

(39) The methods and compositions described below provide combinations of multiple effectors, or single or multiple effector(s) fused with a PTD, or single or multiple effector(s) fused with a targeting moiety, or single or multiple effector(s) fused with a PTD and a targeting moiety, in a chimeric recombinant protein, along with an emollient compound (such as CAGE) for use as a topical anti-inflammatory crème for many different ailments.

(40) Compositions

(41) Provided herein are compositions comprising engineered bacterial effector polypeptides for use in the treatment of inflammation. The engineered bacterial effector polypeptides can be configured as a set of paired peptides. More specifically, a set of paired peptides can be a construct comprising a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof. The set of paired peptides can be linked to one or more polypeptide sequences that facilitate intracellular delivery of the paired peptides, for example, a protein transduction domain (PTD) or a cell penetrating peptide (CPP).

(42) Bacterial effector polypeptides. The first bacterial effector polypeptide and the second bacterial effector polypeptide can be a bacterial effector polypeptide selected from the exemplary bacterial effector polypeptides from a variety of bacteria as shown in Tables 1 and 2. Their enzymatic activity and host targets are also shown in Tables 1 and 2. Representative Uniprot or Genbank references for the polypeptides are shown in Tables 3 and 4. Additional amino acid sequences for, and nucleic acid sequences encoding, these bacterial effector polypeptides can be identified from databases such as UniProt, NCBI, GenBank and publications extant in the art.

(43) TABLE-US-00001 TABLE 1 BACTERIAL T3SS EFFECTORS Effector Bacteria Intracellular Activity Host Target OspF *Shigella* Phosphothreonine ERK, p38 MAPKs *flexneri* lyase OspG *Shigella* Serine/threonine E2 ubiquitin ligases *flexneri* kinase NleH1 EPEC-EHEC Serine/threonine kinase RPS3 NleE/OspZ EPEC-EHEC/ Cysteine methylase TAB2/NfKB *Shigella* NleB EPEC-EHEC O-GlcNAc transferase FADD, GAPDH, RIPK1, TRADD NleC EPEC-EHEC Zinc metalloprotease NFkB YopH *Yersinia* Phosphotyrosine Akt/FAK phosphatase YopE *Yersinia* Rho GAP Rho GTPases/ caspases YopP/YopJ *Yersinia* Acetyltransferase MAPKs YopM *Yersinia* LRR motif PKN/RSK

(44) TABLE-US-00002 TABLE 2 BACTERIAL T3SS E3 UBIQUITIN LIGASE EFFECTORS E3 Ligase Factor Type/ Intracellular Host Family Factor Bacteria Activity Target HECT SopA *Salmonella* Regulation of host TRIM65/56 *typhimurium* inflammation NleL EPEC/EHEC Formation of actin Unknown pedestal RING NleG EPEC/EHEC Unknown Unknown U-Box LubX *Legionella* Regulation of another Cdh1, SidH *pneumophila* effector function Gob X *L. pneumophila* Unknown Unknown NEL IpaH1A *Shigella* Inhibition of NF-Kb HOIP *flexneri* activation IpaH2.5 *S. flexneri* Inhibition of NF-kB HOIP activation IaH4.5 *S. flexneri* Inhibition of NF-kB p65, TBK1 and I-IFN activation IpaH7.8 *S. flexneri* Induction of GLMN pyroptosis

IpaH9.8 *S. flexneri* Inhibition of NF-kB NEMO activation IpaH0722 *S. flexneri* Inhibition of NF-kB TRAF2 activation SspH1 *S. typhimurium* Inhibition of androgen PKN1 receptor SspH2 *S. typhimurium* Promotion of IL-8 Nod1, secretion SGT1 SlrP *S. typhimurium* Induction of host cell Trx death SidE SidC *L. pneumophila* Unknown Unknown family (SdcA)

(45) TABLE-US-00003 TABLE 3 Representative Amino Acid Sequences of Bacterial T3SS Effectors UniProt or Genbank Effector Bacteria Host Target Reference ID OspF *Shigella* ERK, p38 MAPKs Q8VSP9 *flexneri* (OSPF_SHIFL) OspG *Shigella* E2 ubiquitin ligases Q99PZ6 *flexneri* (OSPG_SHIFL) NleH1 EPEC-EHEC RPS3 Q8X831 (Q8X831_ECO57) NleE/OspZ EPEC-EHEC/TAB2/NfKB Q7DBA6 *Shigella* (Q7DBA6_ECO57) NleB EPEC-EHEC FADD, GAPDH, VEC94465.1 RIPK1, TRADD (Genbank) NleC EPEC-EHEC NFkB CBG88408.1 (Genbank) YopH *Yersinia* Akt/FAK P15273 (YOPH_YEREN) YopE *Yersinia* Rho P31492 GTPases/caspases (YOPE_YEREN) YopP/YopJ *Yersinia* MAPKs O34336 (YOPP_BACSU) YopM *Yersinia* PKN/RSK P17778 (YOPM_YERPE)

(46) TABLE-US-00004 TABLE 4 Representative Amino Acid Sequences of Bacterial T3ss E3 Ubiquitin Ligase Effectors E3 Ligase Factor Type/ Host UniProt or Genbank Family Factor Bacteria Target Reference ID HECT SopA *Salmonella* TRIM65/ Q8ZNR3 *typhimurium* 56 (SOPA_SALTY) NleL EPEC/EHEC Unknown A0A0D6ZN92 (A0A0D6ZN92_ECOLX) RING NleG EPEC/EHEC Unknown A0A023YUN6 (A0A023YUN6_ECOLX) U-Box LubX *Legionella* Cdh1, Q5ZRQ0 *pneumophila* SidH (LUBX_LEGPH) Gob X *L. Unknown pneumophila* NEL IpaH1.4 *Shigella* HOIP A0A380D0I4 *flexneri* (A0A380D0I4_SHIFL) IpaH2.5 *S. flexneri* HOIP Q99Q42 (Q99Q42_SHIFL) IaH4.5 *S. flexneri* p65, TBK1 P18009 (IPA4_SHIFL) IpaH7.8 *S. flexneri* GLMN P18014 (IPA7_SHIFL) IpaH9.8 *S. flexneri* NEMO Q8VSC3 (IPA9_SHIFL) IpaH0722 *S. flexneri* TRAF2 SspH1 *S. PKN1* D0ZVG2 *typhimurium* (SSPH1_SALT1) SspH2 *S. Nod1*, P0CE12 *typhimurium* SGT1 (SSPH2_SALTY) SlrP *S. Trx* Q8ZQQ2 *typhimurium* (SLRP_SALTY) SidE SidC *L. Unknown* Q6RCR3 family (SdcA) *pneumophila* (Q6RCR3_LEGPN)

(47) Thus a bacterial effector peptide can be an SspH1; SspH2; SlrP, IpaH1.4, IpaH2.5, IpaH3, IpaH4.5, IpaH7.8, IpaH9.8, NleE, NleC; NleD; NleB; NleH, NleH1, YopM; YopE; YopH; YopJ; YopP; OspG; OspF; OspZ; OspI, SopE; SopB; SopE2; SipA; AvrA; SseL; EspT; or a TiR polypeptide.

(48) In some embodiments, the bacterial effector polypeptide can have an amino acid sequence at least 90% identical to an amino acid sequence set forth in any of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the bacterial effector polypeptide have an amino acid sequence as set forth in any of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79.

(49) The term “type III secretion system or T3SS” refers to a highly specialized molecular needle construct containing a Yersinia injectisome spanning the bacterial membranes, *Yersinia* outer protein (Yop) effectors and Yop translocators needed to deliver the effectors across the membrane (Camelis G R., Int J Med Microbial. 2002 February; 291(6-7):455-62). Pathogenic Yersinia require this T3SS to survive and replicate extracellularly within lymphoid tissues of their animal or human hosts. See, also, U.S. Pat. No. 8,840,901.

(50) The term “immunomodulatory effector protein” refers to small proteins, generally bacterial in origin, that suppress the human innate immune system during infection. These effector proteins activate the NFkB, JNK, and p38 signaling pathways during infection. These effector proteins are often secreted into the cells targeted for infection by T3SS. Once inside the cell, each effector protein targets a single host protein required for innate immunity, which it inactivates using a myriad of mechanisms including acetylation, methylation, action of phosphatases on P04 proteins, induced protein degradation etc.

(51) All T3SS bacterial effector proteins are very small, globular, highly stable, highly catalytic, have high substrate specificity, bind co-factors very tightly, and can be injected into the cell in

denatured form. They modify every substrate molecule in the cell. For instance, the EPEC effector NleE is a cysteine methyltransferase which has a single target in the cell, i.e., the TAB2 scaffold protein in the NFkB pathway. EPEC are mildly infective gut bacteria that attach to colon cells and directly inject virulence proteins thru a T3SS to control innate immune pathways as a survival strategy. During EPEC infection, the injected NleE protein methylates every molecule of cellular TAB2; thereby completely shutting off NFkB signaling, as described in Yao, Q. et al., Structure and Specificity of the Bacterial Cysteine Methyltransferase Effector NleE Suggests a Novel Substrate in Human DNA Repair Pathway., PLoS Pathogens (November 2014) 10(11):e1004522; doi:10.1371/journal.ppat.1004522.

(52) Still another effector protein is YopM, discussed in U.S. Pat. No. 8,840,901 and Rüter, C & Hardwidge, P R, 'Drugs from Bugs': bacterial effector proteins as promising biological (immune-) therapeutics. FEMS Microbiol Lett 351 (December 2013/January 2014) 126-132.

(53) In some embodiments, the first and second bacterial effector polypeptides are different. For example, they can have different amino acid sequences, different structures, different functions, different molecular targets, or have non-overlapping redundant roles in inhibiting an inflammatory pathway, for example, the NFkB, JNK, p38, and STING pathways.

(54) Also included as effectors or effector proteins of the compositions and methods described herein are functional equivalents of the proteins described above. By the term "functional equivalent" is meant any amino acid sequence or modification thereof that has the same targeting and immune suppressing function of the naturally occurring effector protein. In one embodiment, such functional equivalents can have modifications of one or more amino acids from the known sequences. In one embodiment, such functional equivalents can be a smaller fragment of the known sequences. In one embodiment, such functional equivalents can be a derivative of the naturally occurring sequences or be derived from other than human sources. In one embodiment, such functional equivalents can be altered by chemical modification or be altered by recombinant production to be associated with sequences with which the effector proteins are not associated in nature. Similarly, chemical or structural changes or fragments of the nucleic acid sequences that encode the effector proteins are also considered functional equivalents herein.

(55) The paired peptides can be joined by a linker. A linker can be any reagent, molecule or macromolecule that connects the first and second bacterial effector polypeptides such that the linker does not substantially alter the physiological activity of the effector polypeptides. A linker can be a peptide bond. That is, the first and second bacterial effector polypeptides or fragments thereof can be a fusion polypeptide comprising one or more amino acid segments from the first bacterial effector polypeptide and one or more amino acid segments from second bacterial effector polypeptide. The term "amino acid segment" as used herein refers to a contiguous stretch of amino acids within a polypeptide. For example, the amino acid residues 30 to 40 within a 100 amino acid polypeptide would be considered an amino acid segment. An amino acid segment can be a length greater than eight amino acid residues (e.g., greater than about nine, ten, 15, 20, 25, 30, 40, 50, 75, 100, 150, 200, 500, 1000, or more amino acid residues). In some embodiments, an amino acid segment can have a length less than 1000 amino acid residues (e.g., less than 500, less than 400, less than 350, less than 300, less than 200, or less than 100 amino acid residues). In other embodiments, an amino acid segment can have a length from about 20 to about 200 amino acid residues (e.g., about 30 to about 180 amino acid residues, or about 40 to about 150 amino acid residues).

(56) The amino acid segments of the first bacterial effector polypeptide can be contiguous with the amino acid segments of the second or they can be separated by amino acids inserted as a structural spacer. A spacer segment can be one or more amino acids. The one or more amino acids can include amino acids that are the same or that are different. For example, a spacer can be a repeating series of a neutral amino acid (e.g., glycine, alanine, valine, isoleucine or leucine) ranging in number from 1 to 10 or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more). Another example of a spacer

configuration can be a series of interspersed amino acids that may be neutral (e.g., glycine-alanine-glycine-alanine-glycine- alanine, or glycine-glycine-glycine-valine-valine-valine) or charged amino acids (e.g., glutamate- glutamate-glutamate-arginine-arginine-arginine, or aspartate-lysine-aspartate-lysine-aspartate-lysine) or amino acids with other functional groups (e.g., proline-proline-proline-serine-serine-serine or tyrosine-glutamine-cysteine-methionine-tryptophan) ranging in number from 1 to 10 or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more). In another embodiment, a spacer configuration can be a sequence of amino acids derived from a naturally occurring protein such as the hinge region joining the heavy chain CH1 and CH2 domains of immunoglobulin G. In some embodiments, the linker can be a GSGS linker (SEQ ID NO. 92) or a PAPA linker (SEQ ID NO. 91). In some embodiments, the fusion protein comprising a set of paired peptides can exclude a linker.

(57) A fusion protein can be produced in vitro by continuous peptide synthesis according to standard chemical methods known to those in the art. Synthetic polypeptides can also be purchased from commercial sources. A fusion protein can also be produced by recombinant DNA techniques. Nucleic acid segments encoding the first bacterial effector polypeptide can be operably linked in the same open reading frame to nucleic acid sequences encoding the second bacterial effector polypeptide in a vector that includes the requisite regulatory elements, e.g., promoter sequences, transcription initiation sequences, and enhancer sequences, for expression in prokaryotic or eukaryotic cells.

(58) The paired peptide constructs can include a combination of any of an SspH1; SspH2; SlrP; IpaH1.4; IpaH2.5; IpaH3; IpaH4.5, IpaH7.8; IpaH9.8, NleE, NleC; NleD; NleB; NleH, NleH1; YopM; YopE; YopH; YopJ; YopP; OspG; OspF; OspZ; OspI; SopE; SopB; SopE2; SipA; AvrA; SseL; EspT; or a TiR polypeptide. Thus, the first bacterial effector polypeptide can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP.

(59) In some embodiments, the paired peptide construct can include a first bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof or an NLeE polypeptide or a fragment thereof and a second bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof or an NLeE polypeptide or a fragment thereof. In some embodiments, the first bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof is an NLeE polypeptide or a fragment thereof.

(60) In some embodiments, the paired peptide constructs can be configured as summarized in Table 5 below.

(61) TABLE-US-00005 TABLE 5 Paired Peptide Fusion Constructs

Construct name	Effector 1	Linker	Effector 2	SEQ ID NO
YopM PTD- YopM (L-rich) — EPEC	NLeE 80	NleE (No linker)		
YopM PTD- YopM (L-rich) GSGS EPEC	NLeE 82	NleE (GSGS linker)		
YopM PTD- YopM (L-rich) PAPA EPEC	NLeE 81	NleE (PAPA linker)		

(62) In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 85% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 90% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 95% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 99% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, the paired peptide fusion protein can have an amino acid sequence as set forth in

SEQ ID NOs. 80, 81, 82, 83, 84, 85, 86, 87, or 88.

(63) Protein transduction domains. The set of paired peptides can be linked to one or more polypeptide sequences that facilitate intracellular delivery of the paired peptide. The terms “protein transduction domain (PTDs)” and/or “cell-penetrating peptide (CPPs)” refers to powerful sequences that allow intracellular delivery of conjugated cargoes to modify cell behavior. These small peptides can transport a wide variety of biologically active conjugates into the cell. Heterologous CPP coding sequences are added to effectors or effector-fusions to facilitate cellular uptake of the proteins into cells and tissues, including use of endogenous CPPs encoded in native effector proteins. This includes addition of CPP sequences or modules to the effector via chemical crosslinking, attachment to a nano-particle or other scaffold chemically or via PPIs for the purpose of transporting the effector across tissue and cell membranes. Among useful PTD or CPPs for the present methods and compositions are those known and identified in the art, including, without limitation, HIV Tat protein basic domain, (HIV Tat amino acids 48-60 or 49-57), poly-Arg or polyLys, penetratin, MPG, Pep-1, MAP, and transportan. See, e.g., Table 1 of Guidotti, G. et al, Trends in Pharmacological Sciences (April 2017), 38(4):406-424, which includes additional examples of CPPs and sequences origins and properties. Other CPPs are described in Norkowski, S. et al, Bacterial LPX motif-harboring virulence factors constitute a species-spanning family of cell-penetrating effectors, Cellular and Molecular Life Sciences (December 2017) doi.org/10.1007/s00018-017-2733-4, which described prototypes of such bacteria-derived cell-penetrating effectors (CPEs) including the *Yersinia enterocolitica*-derived YopM, the *Salmonella typhimurium* effector SspH1, and the *Shigella* IpaH proteins. Still other protein transporter molecules include those previously described in Dixon, J E et al, Proc. Natl Acad Sci, (January 2016), E291-299; as well as synthetic protein mimics described by Tezgel, A O et al BioMacromolecules (2017) 16:819-825. See, also, Bolhassani, A. et al, In vitro and in vivo delivery of therapeutic proteins using cell penetrating peptides. Peptides (November 2016), 87:50-63, which discusses useful CPPs for the present compositions and methods, including without limitation, covalent bonded CPPs, such as Poly-Arg peptides, Tat and VP22, df Tat, Cyclic CPPs, IMT-P8 (particularly useful for transdermal delivery), seven arginine (R7) and Streptolysin O (SLO)-mediated systems and elastin like polypeptide, CPP-adaptor system, 1,2-Benzisothiazolin-3-one (BIT) and Tat, activatable cell-penetrating peptides, LDP12, M918, BR2, peptide for ocular delivery (POD), native protein independent of R11-CPP, Poly-arginine/Tat and Tat-PTD among others. Also identified are non-covalent bonded CPPs such as Pep-1, CADY-2, R8 and azo-R8, Penetratin, HR9 and IR9 peptides and pVEC. All of these documents are incorporated by reference herein for detailed descriptions of known CPPs and PTDs. It is also anticipated that novel PTD/CPPs will prove useful with the compositions described herein.

(64) Exemplary protein transduction domains include a YopM protein transduction domain, an SspH1 protein transduction domain, or an IpaH protein transduction domain. A useful YopM protein transduction domain can have an amino acid sequence as set forth in SEQ ID NO 5.

(65) In some embodiments, a fusion protein comprising a set of paired peptides linked to a protein transduction domain can have an amino acid sequence is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 90% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 95% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 99% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, the paired peptide fusion protein can have an amino acid sequence as set forth in SEQ ID NOs. 10, 13, 16, 19, 22, or 24.

(66) Polypeptides. We tend to use the term “protein” to refer to longer or larger amino acid polymers, and we tend to use the term “polypeptide” to refer to shorter sequences or to a chain of

amino acid residues within a larger molecule (e.g., within a fusion protein) or complex. Both terms, however, are meant to describe an entity of two or more subunit amino acids, amino acid analogs, or other peptidomimetics, regardless of post-translational modification (e.g., amidation, phosphorylation or glycosylation). The subunits can be linked by peptide bonds or other bonds such as, for example, dicysteine, ester or ether bonds. The terms “amino acid” and “amino acid residue” refer to natural and/or unnatural or synthetic amino acids, which may be D- or L-form optical isomers. Full-length proteins, analogs, mutants, and fragments thereof are encompassed by this definition.

(67) The amino acid sequence of the bacterial effector polypeptides disclosed herein can be identical to the wild-type sequences of appropriate components. Alternatively, any of the components can contain mutations such as deletions, additions, or substitutions. All that is required is that the variant bacterial effector polypeptide have at least 5% (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99%, 100%, or even more) of the ability of the bacterial effector polypeptide containing only wild-type sequences to specifically bind the target. Substitutions will preferably be conservative substitutions. Conservative substitutions typically include substitutions within the following groups: glycine and alanine; valine, isoleucine, and leucine; aspartic acid and glutamic acid; asparagine, glutamine, serine and threonine; lysine, histidine and arginine; and phenylalanine and tyrosine.

(68) Variant bacterial effector polypeptides, e.g., those having one or more amino acid substitutions relative to a native bacterial effector polypeptide amino acid sequence, can be prepared and modified as described herein. Amino acid substitutions can be made, in some cases, by selecting substitutions that do not differ significantly in their effect on maintaining (a) the structure of the peptide backbone in the area of the substitution, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. For example, naturally occurring residues can be divided into groups based on side-chain properties: (1) hydrophobic amino acids (norleucine, methionine, alanine, valine, leucine, and isoleucine); (2) neutral hydrophilic amino acids (cysteine, serine, and threonine); (3) acidic amino acids (aspartic acid and glutamic acid); (4) basic amino acids (asparagine, glutamine, histidine, lysine, and arginine); (5) amino acids that influence chain orientation (glycine and proline); and (6) aromatic amino acids (tryptophan, tyrosine, and phenylalanine). Substitutions made within these groups can be considered conservative substitutions. Non-limiting examples of useful substitutions include, without limitation, substitution of valine for alanine, lysine for arginine, glutamine for asparagine, glutamic acid for aspartic acid, serine for cysteine, asparagine for glutamine, aspartic acid for glutamic acid, proline for glycine, arginine for histidine, leucine for isoleucine, isoleucine for leucine, arginine for lysine, leucine for methionine, leucine for phenylalanine, glycine for proline, threonine for serine, serine for threonine, tyrosine for tryptophan, phenylalanine for tyrosine, and/or leucine for valine. Variant bacterial effector polypeptides having conservative and/or non-conservative substitutions (e.g., with respect to any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24), as well as fragments of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, fragments of variants of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88 and polypeptides comprising any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, variants or fragments of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, or fragments of variants of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, can be screened for biological activity using suitable assays, including those described herein. For example, the activity of a bacterial effector polypeptide, for example,

NLeE or a mutant or fragment thereof, can be evaluated *in vitro* by assaying for methylase activity or in cell based systems to characterize its effect on cytokine release.

(69) In some embodiments, a bacterial effector polypeptide can comprise an amino acid sequence as set forth in SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, but with a particular number of amino acid substitutions. For example, a bacterial effector polypeptides can have the amino acid sequence of any one of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, but with one, two, three, four, or five amino acid substitutions.

(70) In some embodiments, a bacterial effector polypeptide as provided herein can include an amino acid sequence with at least 85% (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 97.5%, 98%, 98.5%, 99.0%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) sequence identity to a region of a reference bacterial effector polypeptide sequence (e.g., SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88). Methods of determining percent sequence identity are discussed below.

(71) In some embodiments, a polypeptide provided herein can be a substantially pure polypeptide. As used herein, the term “substantially pure” with reference to a polypeptide means that the polypeptide is substantially free of other polypeptides, lipids, carbohydrates, and nucleic acid with which it is naturally associated. Thus, a substantially pure polypeptide is any polypeptide that is removed from its natural environment and is at least 60 percent pure or is any chemically synthesized polypeptide. A substantially pure polypeptide can be at least about 60, 65, 70, 75, 80, 85, 90, 95, or 99 percent pure. Typically, a substantially pure polypeptide will yield a single major band on a non-reducing polyacrylamide gel.

(72) A variety of methods can be used to make a polypeptide including, for example, expression by prokaryotic systems, expression by eukaryotic systems, and chemical synthesis techniques. Exemplary methods for polypeptide purification purificinclud, without limitation, fractionation, centrifugation, and chromatography, e.g., gel filtration, ion exchange chromatography, reverse-phase HPLC and immunoaffinity purification.

(73) A polypeptide can be modified by linkage to a polymer such as polyethylene glycol (PEG), or by fusion to another polypeptide such as albumin, for example. For example, one or more PEG moieties can be conjugated to a bacterial effector polypeptide or fusion protein via lysine residues. Linkage to PEG or another suitable polymer, or fusion to albumin or another suitable polypeptide can result in a modified bacterial effector polypeptide or fusion protein having an increased half life as compared to an unmodified bacterial effector polypeptide or fusion protein. Without being bound by a particular mechanism, an increased serum half life can result from reduced proteolytic degradation, immune recognition, or cell scavenging of the modified bacterial effector polypeptide or fusion protein. Methods for modifying a polypeptide by linkage to PEG (also referred to as “PEGylation”) or other polymers include those set forth in U.S. Pat. No. 6,884,780; Cataliotti et al. ((2007) Trends Cardiovasc. Med. 17:10-14; Veronese and Mero (2008) BioDrugs 22:315-329; Miller et al. (2006) Bioconjugate Chem. 17:267-274; and Veronese and Pasut (2005) Drug Discov. Today 10:1451-1458, all of which are incorporated herein by reference in their entirety. Methods for modifying a polypeptide by fusion to albumin include those set forth in U.S. Patent Publication No. 20040086976, and Wang et al. (2004) Pharm. Res. 21:2105-2111, both of which are incorporated herein by reference in their entirety.

(74) Nucleic acids. We may use the terms “nucleic acid” and “polynucleotide” interchangeably to refer to both RNA and DNA, including cDNA, genomic DNA, synthetic DNA, and DNA (or RNA) containing nucleic acid analogs, any of which may encode a polypeptide of the invention and all of which are encompassed by the invention. Polynucleotides can have essentially any three-dimensional structure. A nucleic acid can be double-stranded or single-stranded (i.e., a sense strand

or an antisense strand). Non-limiting examples of polynucleotides include genes, gene fragments, exons, introns, messenger RNA (mRNA) and portions thereof, transfer RNA, ribosomal RNA, siRNA, micro-RNA, ribozymes, cDNA, recombinant polynucleotides, branched polynucleotides, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, and primers, as well as nucleic acid analogs. In the context of the present invention, nucleic acids can encode a bacterial effector polypeptide, paired peptide fusion protein, or construct comprising a paired peptide fusion protein linked to a protein transduction domain.

(75) An “isolated” nucleic acid can be, for example, a naturally-occurring DNA molecule or a fragment thereof, provided that at least one of the nucleic acid sequences normally found immediately flanking that DNA molecule in a naturally-occurring genome is removed or absent. Thus, an isolated nucleic acid includes, without limitation, a DNA molecule that exists as a separate molecule, independent of other sequences (e.g., a chemically synthesized nucleic acid, or a cDNA or genomic DNA fragment produced by the polymerase chain reaction (PCR) or restriction endonuclease treatment). An isolated nucleic acid also refers to a DNA molecule that is incorporated into a vector, an autonomously replicating plasmid, a virus, or into the genomic DNA of a prokaryote or eukaryote. In addition, an isolated nucleic acid can include an engineered nucleic acid such as a DNA molecule that is part of a hybrid or fusion nucleic acid. A nucleic acid existing among many (e.g., dozens, or hundreds to millions) of other nucleic acids within, for example, cDNA libraries or genomic libraries, or gel slices containing a genomic DNA restriction digest, is not an isolated nucleic acid.

(76) Isolated nucleic acid molecules can be produced by in several ways. For example, polymerase chain reaction (PCR) techniques can be used to obtain an isolated nucleic acid containing a nucleotide sequence described herein, including nucleotide sequences encoding a polypeptide described herein. PCR can be used to amplify specific sequences from DNA as well as RNA, including sequences from total genomic DNA or total cellular RNA. Generally, sequence information from the ends of the region of interest or beyond is employed to design oligonucleotide primers that are identical or similar in sequence to opposite strands of the template to be amplified. Various PCR strategies also are available by which site-specific nucleotide sequence modifications can be introduced into a template nucleic acid.

(77) Isolated nucleic acids also can be chemically synthesized, either as a single nucleic acid molecule (e.g., using automated DNA synthesis in the 3' to 5' direction using phosphoramidite technology) or as a series of oligonucleotides. For example, one or more pairs of long oligonucleotides (e.g., >50-100 nucleotides) can be synthesized that contain the desired sequence, with each pair containing a short segment of complementarity (e.g., about 15 nucleotides) such that a duplex is formed when the oligonucleotide pair is annealed. DNA polymerase is used to extend the oligonucleotides, resulting in a single, double-stranded nucleic acid molecule per oligonucleotide pair, which then can be ligated into a vector.

(78) Two nucleic acids or the polypeptides they encode may be described as having a certain degree of identity to one another. For example, a bacterial effector polypeptide and a biologically active variant thereof may be described as exhibiting a certain degree of identity. Alignments may be assembled by locating short sequences in the Protein Information Research (PIR) site (<http://pir.georgetown.edu>), followed by analysis with the “short nearly identical sequences” Basic Local Alignment Search Tool (BLAST) algorithm on the NCBI website (<http://www.ncbi.nlm.nih.gov/blast>).

(79) As used herein, the term “percent sequence identity” refers to the degree of identity between any given query sequence and a subject sequence. For example, a bacterial effector polypeptide disclosed herein can be the query sequence and a fragment of a bacterial effector polypeptide can be the subject sequence. Similarly, a fragment of bacterial effector polypeptide can be the query sequence and a biologically active variant thereof can be the subject sequence.

(80) To determine sequence identity, a query nucleic acid or amino acid sequence can be aligned to

one or more subject nucleic acid or amino acid sequences, respectively, using the computer program ClustalW (version 1.83, default parameters), which allows alignments of nucleic acid or protein sequences to be carried out across their entire length (global alignment).

(81) ClustalW calculates the best match between a query and one or more subject sequences and aligns them so that identities, similarities and differences can be determined. Gaps of one or more residues can be inserted into a query sequence, a subject sequence, or both, to maximize sequence alignments. For fast pair wise alignment of nucleic acid sequences, the following default parameters are used: word size: 2; window size: 4; scoring method: percentage; number of top diagonals: 4; and gap penalty: 5. For multiple alignments of nucleic acid sequences, the following parameters are used: gap opening penalty: 10.0; gap extension penalty: 5.0; and weight transitions: yes. For fast pair wise alignment of protein sequences, the following parameters are used: word size: 1; window size: 5; scoring method: percentage; number of top diagonals: 5; gap penalty: 3. For multiple alignment of protein sequences, the following parameters are used: weight matrix: blosum; gap opening penalty: 10.0; gap extension penalty: 0.05; hydrophilic gaps: on; hydrophilic residues: Gly, Pro, Ser, Asn, Asp, Gln, Glu, Arg, and Lys; residue-specific gap penalties: on. The output is a sequence alignment that reflects the relationship between sequences. ClustalW can be run, for example, at the Baylor College of Medicine Search Launcher site (searchlauncher.bcm.tmc.edu/multi-align/multi-align.html) and at the European Bioinformatics Institute site on the World Wide Web (ebi.ac.uk/clustalw).

(82) To determine a percent identity between a query sequence and a subject sequence, ClustalW divides the number of identities in the best alignment by the number of residues compared (gap positions are excluded), and multiplies the result by 100. The output is the percent identity of the subject sequence with respect to the query sequence. It is noted that the percent identity value can be rounded to the nearest tenth. For example, 78.11, 78.12, 78.13, and 78.14 are rounded down to 78.1, while 78.15, 78.16, 78.17, 78.18, and 78.19 are rounded up to 78.2.

(83) The nucleic acids and polypeptides described herein may be referred to as “exogenous”. The term “exogenous” indicates that the nucleic acid or polypeptide is part of, or encoded by, a recombinant nucleic acid construct, or is not in its natural environment. For example, an exogenous nucleic acid can be a sequence from one species introduced into another species, i.e., a heterologous nucleic acid. Typically, such an exogenous nucleic acid is introduced into the other species via a recombinant nucleic acid construct. An exogenous nucleic acid can also be a sequence that is native to an organism and that has been reintroduced into cells of that organism. An exogenous nucleic acid that includes a native sequence can often be distinguished from the naturally occurring sequence by the presence of non-natural sequences linked to the exogenous nucleic acid, e.g., non-native regulatory sequences flanking a native sequence in a recombinant nucleic acid construct. In addition, stably transformed exogenous nucleic acids typically are integrated at positions other than the position where the native sequence is found.

(84) Nucleic acids of the invention, that is, nucleic acids having a nucleotide sequence of any paired peptides fusion proteins and constructs disclosed herein, can include nucleic acids sequences that are at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 99% identical to the reference sequences disclosed herein.

(85) A nucleic acid, i.e., an oligonucleotide (e.g., a probe or a primer) that is specific for a target nucleic acid will hybridize to the target nucleic acid under suitable conditions. We may refer to hybridization or hybridizing as the process by which an oligonucleotide single strand anneals with a complementary strand through base pairing under defined hybridization conditions. It is a specific, i.e., non-random, interaction between two complementary polynucleotides. Hybridization and the strength of hybridization (i.e., the strength of the association between the nucleic acids) is influenced by such factors as the degree of complementarity between the nucleic acids, stringency of the conditions involved, and the melting temperature (T_m) of the formed hybrid. The hybridization products can be duplexes or triplexes formed with targets in solution or on solid supports.

(86) Vectors. Vectors containing nucleic acids such as those described herein also are provided. A “vector” is a replicon, such as a plasmid, phage, or cosmid, into which another DNA segment may be inserted so as to bring about the replication of the inserted segment. Generally, a vector is capable of replication when associated with the proper control elements. Suitable vector backbones include, for example, those routinely used in the art such as plasmids, viruses, artificial chromosomes, BACs, YACs, or PACs. The term “vector” includes cloning and expression vectors, as well as viral vectors and integrating vectors. An “expression vector” is a vector that includes a regulatory region. A wide variety of host/expression vector combinations may be used to express the nucleic acid sequences described herein. Suitable expression vectors include, without limitation, plasmids and viral vectors derived from, for example, bacteriophage, baculoviruses, and retroviruses. Numerous vectors and expression systems are commercially available from such corporations as Novagen (Madison, Wis.), Clontech (Palo Alto, Calif.), Stratagene (La Jolla, Calif.), and Invitrogen/Life Technologies (Carlsbad, Calif.).

(87) The vectors provided herein also can include, for example, origins of replication, scaffold attachment regions (SARs), and/or markers. A marker gene can confer a selectable phenotype on a host cell. For example, a marker can confer biocide resistance, such as resistance to an antibiotic (e.g., kanamycin, G418, bleomycin, or hygromycin). As noted above, an expression vector can include a tag sequence designed to facilitate manipulation or detection (e.g., purification or localization) of the expressed polypeptide. Tag sequences, such as green fluorescent protein (GFP), glutathione S-transferase (GST), polyhistidine, c-myc, hemagglutinin, or Flag™ tag sequences typically are expressed as a fusion with the encoded polypeptide. Such tags can be inserted anywhere within the polypeptide, including at either the carboxyl or amino terminus.

(88) Additional expression vectors also can include, for example, segments of chromosomal, non-chromosomal and synthetic DNA sequences. Suitable vectors include derivatives of SV40 and known bacterial plasmids, e.g., *E. coli* plasmids col E1, pCR1, pBR322, pMal-C2, pET, pGEX, pMB9 and their derivatives, plasmids such as RP4; phage DNAs, e.g., the numerous derivatives of phage 1, e.g., NM989, and other phage DNA, e.g., M13 and filamentous single stranded phage DNA; yeast plasmids such as the 2p plasmid or derivatives thereof, vectors useful in eukaryotic cells, such as vectors useful in insect or mammalian cells; vectors derived from combinations of plasmids and phage DNAs, such as plasmids that have been modified to employ phage DNA or other expression control sequences.

(89) Yeast expression systems can also be used. For example, the non-fusion pYES2 vector (XbaI, SphI, ShoI, NotI, GstXI, EcoRI, BstXI, BamH1, SacI, KpnI, and HindIII cloning sites; Invitrogen) or the fusion pYESHisA, B, C (XbaI, SphI, ShoI, NotI, BstXI, EcoRI, BamH1, SacI, KpnI, and HindIII cloning sites).

(90) The vector can also include a regulatory region. The term “regulatory region” refers to nucleotide sequences that influence transcription or translation initiation and rate, and stability and/or mobility of a transcription or translation product. Regulatory regions include, without limitation, promoter sequences, enhancer sequences, response elements, protein recognition sites, inducible elements, protein binding sequences, 5′ and 3′ untranslated regions (UTRs), transcriptional start sites, termination sequences, polyadenylation sequences, nuclear localization signals, and introns.

(91) As used herein, the term “operably linked” refers to positioning of a regulatory region and a sequence to be transcribed in a nucleic acid so as to influence transcription or translation of such a sequence. For example, to bring a coding sequence under the control of a promoter, the translation initiation site of the translational reading frame of the polypeptide is typically positioned between one and about fifty nucleotides downstream of the promoter. A promoter can, however, be positioned as much as about 5,000 nucleotides upstream of the translation initiation site or about 2,000 nucleotides upstream of the transcription start site. A promoter typically comprises at least a core (basal) promoter. A promoter also may include at least one control element, such as an

enhancer sequence, an upstream element or an upstream activation region (UAR). The choice of promoters to be included depends upon several factors, including, but not limited to, efficiency, selectability, inducibility, desired expression level, and cell- or tissue-preferential expression. (92) Vectors include, for example, viral vectors (such as adenoviruses ("Ad"), adeno-associated viruses (AAV), and vesicular stomatitis virus (VSV) and retroviruses), liposomes and other lipid-containing complexes, and other macromolecular complexes capable of mediating delivery of a polynucleotide to a host cell. Vectors can also comprise other components or functionalities that further modulate gene delivery and/or gene expression, or that otherwise provide beneficial properties to the targeted cells.

(93) A "recombinant viral vector" refers to a viral vector comprising one or more heterologous gene products or sequences. Since many viral vectors exhibit size-constraints associated with packaging, the heterologous gene products or sequences are typically introduced by replacing one or more portions of the viral genome. Such viruses may become replication-defective, requiring the deleted function(s) to be provided in trans during viral replication and encapsidation (by using, e.g., a helper virus or a packaging cell line carrying gene products necessary for replication and/or encapsidation).

(94) Suitable nucleic acid delivery systems include recombinant viral vector, typically sequence from at least one of an adenovirus, adenovirus-associated virus (AAV), helper-dependent adenovirus, retrovirus, or hemagglutinating virus of Japan-liposome (HVJ) complex. In such cases, the viral vector comprises a strong eukaryotic promoter operably linked to the polynucleotide e.g., a cytomegalovirus (CMV) promoter. The recombinant viral vector can include one or more of the polynucleotides therein, preferably about one polynucleotide. In some embodiments, the viral vector used in the invention methods has a pfu (plaque forming units) of from about 10^{10} to about 5×10^{10} pfu. In embodiments in which the polynucleotide is to be administered with a non-viral vector, use of between from about 0.1 nanograms to about 4000 micrograms will often be useful e.g., about 1 nanogram to about 100 micrograms.

(95) Additional vectors include retroviral vectors such as Moloney murine leukemia viruses and HIV-based viruses. One HIV-based viral vector comprises at least two vectors wherein the gag and pol genes are from an HIV genome and the env gene is from another virus. DNA viral vectors include pox vectors such as orthopox or avipox vectors, herpesvirus vectors such as a herpes simplex I virus (HSV) vector.

(96) Pox viral vectors introduce the gene into the cells cytoplasm. Avipox virus vectors result in only a short term expression of the nucleic acid. Adenovirus vectors, adeno-associated virus vectors and herpes simplex virus (HSV) vectors may be an indication for some invention embodiments. The adenovirus vector results in a shorter term expression (e.g., less than about a month) than adeno-associated virus, in some embodiments, may exhibit much longer expression. The particular vector chosen will depend upon the target cell and the condition being treated. The selection of appropriate promoters can readily be accomplished. An example of a suitable promoter is the 763-base-pair cytomegalovirus (CMV) promoter. Other suitable promoters which may be used for gene expression include, but are not limited to, the Rous sarcoma virus (RSV), the SV40 early promoter region, the herpes thymidine kinase promoter, the regulatory sequences of the metallothionein (MT) gene, prokaryotic expression vectors such as the β -lactamase promoter, the tac promoter, promoter elements from yeast or other fungi such as the Gal 4 promoter, the ADC (alcohol dehydrogenase) promoter, PGK (phosphoglycerol kinase) promoter, alkaline phosphatase promoter; and the animal transcriptional control regions, which exhibit tissue specificity and have been utilized in transgenic animals: elastase I gene control region which is active in pancreatic acinar cells, insulin gene control region which is active in pancreatic beta cells, immunoglobulin gene control region which is active in lymphoid cells, mouse mammary tumor virus control region which is active in testicular, breast, lymphoid and mast cells, albumin gene control region which is active in liver, alpha-fetoprotein gene control region which is active in liver, alpha 1-antitrypsin

gene control region which is active in the liver, beta-globin gene control region which is active in myeloid cells, myelin basic protein gene control region which is active in oligodendrocyte cells in the brain, myosin light chain-2 gene control region which is active in skeletal muscle, and gonadotropic releasing hormone gene control region which is active in the hypothalamus. Certain proteins can be expressed using their native promoter. Other elements that can enhance expression can also be included such as an enhancer or a system that results in high levels of expression such as a *tat* gene and *tar* element. This cassette can then be inserted into a vector, e.g., a plasmid vector such as, pUC19, pUC118, pBR322, or other known plasmid vectors, that includes, for example, an *E. coli* origin of replication. The plasmid vector may also include a selectable marker such as the β -lactamase gene for ampicillin resistance, provided that the marker polypeptide does not adversely affect the metabolism of the organism being treated.

(97) Pharmaceutical carriers. The compositions also include a pharmaceutically acceptable carrier. We use the terms “pharmaceutically acceptable” (or “pharmacologically acceptable”) to refer to molecular entities and compositions that do not produce an adverse, allergic or other untoward reaction when administered to an animal or a human, as appropriate. The term “pharmaceutically acceptable carrier,” as used herein, includes any and all solvents, dispersion media, coatings, antibacterial, isotonic and absorption delaying agents, buffers, excipients, binders, lubricants, gels, surfactants and the like, that may be used as media for a pharmaceutically acceptable substance.

(98) Thus, the invention also includes pharmaceutical compositions which contain, as the active ingredient, a fusion protein comprising a set of paired peptides or a fusion protein comprising a set of paired peptides linked a protein transduction domain, in combination with one or more pharmaceutically acceptable carriers. An active ingredient can be a composition comprising a set of paired peptides linked to a protein transduction domain and wherein prepared peptides comprise a first bacterial effector polypeptide linked to a second bacterial effector polypeptide. In some embodiments, the polypeptide compositions can be sterilized using conventional sterilization techniques before or after it is combined with the pharmaceutically acceptable carrier. In making the compositions of the invention, the polypeptide compositions are typically mixed with an excipient, diluted by an excipient or enclosed within such a carrier in the form of, for example, a capsule, tablet, sachet, paper, or other container. When the excipient serves as a diluent, it can be a solid, semisolid, or liquid material (e.g., normal saline), which acts as a vehicle, carrier or medium for the active ingredient. Thus, the compositions can be in the form of tablets, pills, powders, lozenges, sachets, cachets, elixirs, suspensions, emulsions, solutions, syrups, aerosols (as a solid or in a liquid medium), ointments, soft and hard gelatin capsules, suppositories, sterile injectable solutions, and sterile packaged powders. As is known in the art, the type of diluent can vary depending upon the intended route of administration. The resulting compositions can include additional agents, such as preservatives. The excipient or carrier is selected on the basis of the mode and route of administration. Suitable pharmaceutical carriers, as well as pharmaceutical necessities for use in pharmaceutical formulations, are described in Remington's Pharmaceutical Sciences (E. W. Martin), a well-known reference text in this field, and in the USP/NF (United States Pharmacopeia and the National Formulary). Some examples of suitable excipients include lactose, dextrose, sucrose, sorbitol, mannitol, starches, gum acacia, calcium phosphate, alginates, tragacanth, gelatin, calcium silicate, microcrystalline cellulose, polyvinylpyrrolidone, cellulose, water, syrup, and methyl cellulose. The formulations can additionally include: lubricating agents such as talc, magnesium stearate, and mineral oil; wetting agents; emulsifying and suspending agents; preserving agents such as methyl- and propylhydroxy-benzoates; sweetening agents; and flavoring agents. The pharmaceutical compositions can also be formulated so as to provide quick, sustained or delayed release of the active ingredient after administration to the patient.

(99) Pharmaceutically acceptable compositions for use in the present methods, including those in which the polypeptides are entrapped in a colloid for oral delivery, can be prepared according to standard techniques. The polypeptides can be dried and compacted by grinding or pulverizing and

inserted into a capsule for oral administration. In some embodiments, the polypeptides can be combined one or more excipients, for example, a disintegrant, a filler, a glidant, or a preservative. Suitable capsules include both hard shell capsules or soft-shelled capsules. Any lipid-based or polymer-based colloid may be used to form the capsule. Exemplary polymers useful for colloid preparations include gelatin, plant polysaccharides or their derivatives such as carrageenans and modified forms of starch and cellulose, e.g., hypromellose. Optionally, other ingredients may be added to the gelling agent solution, for example plasticizers such as glycerin and/or sorbitol to decrease the capsule's hardness, coloring agents, preservatives, disintegrants, lubricants and surface treatment. In some embodiments, the capsule does not include gelatin. In other embodiments, the capsule does not include plant polysaccharides or their derivatives.

(100) Regardless of their original source or the manner in which they are obtained, the polypeptides of the invention can be formulated in accordance with their use. These compositions can be prepared in a manner well known in the pharmaceutical art, and can be administered by a variety of routes, depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration may be oral or topical (including ophthalmic and to mucous membranes including intranasal, vaginal and rectal delivery). In some embodiments, administration can be pulmonary (e.g., by inhalation or insufflation of powders or aerosols, including by nebulizer; intratracheal, intranasal, epidermal and transdermal) or ocular. Methods for ocular delivery can include topical administration (eye drops), subconjunctival, periocular or intravitreal injection or introduction by balloon catheter or ophthalmic inserts surgically placed in the conjunctival sac. Parenteral administration includes intravenous, intraarterial, subcutaneous, intraperitoneal or intramuscular injection or infusion; or intracranial, e.g., intrathecal or intraventricular administration. Parenteral administration can be in the form of a single bolus dose, or may be, for example, by a continuous perfusion pump. Pharmaceutical compositions and formulations for topical administration may include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids, powders, and the like. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like may be necessary or desirable.

(101) The compositions can be formulated in a unit dosage form, each dosage containing, for example, from about 0.005 mg to about 2000 mg of polypeptides per daily dose. The term "unit dosage forms" refers to physically discrete units suitable as unitary dosages for human subjects and other mammals, each unit containing a predetermined quantity of active material calculated to produce the desired therapeutic effect, in association with a suitable pharmaceutical excipient. For preparing solid compositions such as tablets, the principal active ingredient is mixed with a pharmaceutical excipient to form a solid preformulation composition containing a homogeneous mixture of a compound of the present invention. When referring to these preformulation compositions as homogeneous, the active ingredient is typically dispersed evenly throughout the composition so that the composition can be readily subdivided into equally effective unit dosage forms such as tablets, pills and capsules. This solid preformulation is then subdivided into unit dosage forms of the type described above containing from, for example, 0.005 mg to about 1000 mg of the compositions of the present invention.

(102) The compositions can be formulated in a unit dosage form, each dosage containing, for example, from about 0.1 mg to about 50 mg, from about 0.1 mg to about 40 mg, from about 0.1 mg to about 20 mg, from about 0.1 mg to about 10 mg, from about 0.2 mg to about 20 mg, from about 0.3 mg to about 15 mg, from about 0.4 mg to about 10 mg, from about 0.5 mg to about 1 mg; from about 0.5 mg to about 100 mg, from about 0.5 mg to about 50 mg, from about 0.5 mg to about 30 mg, from about 0.5 mg to about 20 mg, from about 0.5 mg to about 10 mg, from about 0.5 mg to about 5 mg; from about 1 mg to about 50 mg, from about 1 mg to about 30 mg, from about 1 mg to about 20 mg, from about 1 mg to about 10 mg, from about 1 mg to about 5 mg; from about 5 mg to about 50 mg, from about 5 mg to about 20 mg, from about 5 mg to about 10 mg; from about 10 mg to about 100 mg, from about 20 mg to about 200 mg, from about 30 mg to about 150 mg,

from about 40 mg to about 100 mg, from about 50 mg to about 100 mg of the active ingredient.

(103) In some embodiments, tablets or pills of the present invention can be coated or otherwise compounded to provide a dosage form affording the advantage of prolonged action. For example, the tablet or pill can comprise an inner dosage and an outer dosage component, the latter being in the form of an envelope over the former. The two components can be separated by an enteric layer which serves to resist disintegration in the stomach and permit the inner component to pass intact into the duodenum or to be delayed in release. A variety of materials can be used for such enteric layers or coatings, such materials including a number of polymeric acids and mixtures of polymeric acids with such materials as shellac, cetyl alcohol, and cellulose acetate.

(104) The liquid forms in which the compositions of the present invention can be incorporated for administration orally or by injection include aqueous solutions, suitably flavored syrups, aqueous or oil suspensions, and flavored emulsions with edible oils such as cottonseed oil, sesame oil, coconut oil, or peanut oil, as well as elixirs and similar pharmaceutical vehicles.

(105) The proportion or concentration of the compositions of the invention in a pharmaceutical composition can vary depending upon a number of factors including dosage, chemical characteristics (e.g., hydrophobicity), and the route of administration.

(106) Methods of Treatment

(107) The compositions disclosed herein are generally and variously useful for treatment of inflammatory disorders and conditions. Inflammatory disorders and conditions encompass a wide range of disorders covering many different systems and organs, including the gastrointestinal tract, the skin, the lungs, and the musculoskeletal system. Exemplary inflammatory disorders include inflammatory bowel disease, rheumatoid arthritis, allergy, asthma, autoimmune diseases, coeliac disease, glomerulonephritis, hepatitis, reperfusion injury and transplant rejection. An inflammatory disorder can be a gastrointestinal disorder, for example, inflammatory bowel disease, Crohn's disease, and the ileocolitis, ileocecal, jejunocolitis, and gastroduodenal subtypes of Crohn's disease, and ulcerative colitis and subtypes of ulcerative colitis.

(108) A subject is effectively treated whenever a clinically beneficial result ensues. This may mean, for example, a complete resolution of the symptoms associated with an inflammatory disorder, a decrease in the severity of the symptoms associated with an inflammatory disorder, or a slowing of the progression of symptoms associated with an inflammatory disorder. These methods can further include the steps of a) identifying a subject (e.g., a patient and, more specifically, a human patient) who has an inflammatory disorder; and b) providing to the subject a composition comprising a paired peptide composition disclosed herein in a physiologically acceptable carrier. An amount of such a composition provided to the subject that results in a complete resolution of the symptoms associated with an inflammatory disorder, a decrease in the severity of the symptoms associated with an inflammatory disorder, or a slowing of the progression of symptoms associated with an inflammatory disorder considered a therapeutically effective amount. The present methods may also include a monitoring step to help optimize dosing and scheduling as well as predict outcome.

(109) The methods disclosed herein can be applied to a wide range of species, e.g., humans, non-human primates (e.g., monkeys), horses, pigs, cows or other livestock, dogs, cats or other mammals kept as pets, rats, mice, or other laboratory animals. The compositions described herein are useful in therapeutic compositions and regimens or for the manufacture of a medicament for use in treatment of conditions as described herein (e.g., inflammatory disorders and conditions.)

(110) When formulated as pharmaceuticals, the compositions can be administered to any part of the host's body for subsequent delivery to a target cell. A composition can be delivered to, without limitation, the brain, the cerebrospinal fluid, joints, nasal mucosa, blood, lungs, intestines, muscle tissues, skin, or the peritoneal cavity of a mammal. In terms of routes of delivery, a composition can be administered by intravenous, intracranial, intraperitoneal, intramuscular, subcutaneous, intramuscular, intrarectal, intravaginal, intrathecal, intratracheal, intradermal, or transdermal injection, by oral or nasal administration, or by gradual perfusion over time. In a further example,

an aerosol preparation of a composition can be given to a host by inhalation.

(111) Regardless of how the compositions are formulated, the dosage required will depend on the route of administration, the nature of the formulation, the nature of the subject's condition, e.g., a gastrointestinal disorder or a skin disorder, the subject's size, weight, surface area, age, and sex, other drugs being administered, and the judgment of the attending clinicians. Suitable dosages are in the range of 0.01-1,000 mg/kg. Some typical dose ranges are from about 1 µg/kg to about 1 g/kg of body weight per day. In some embodiments, the dose range is from about 0.01 mg/kg to about 100 mg/kg of body weight per day. In some embodiments, the dose can be, for example, 1 mg/kg, 2 mg/kg, 5 mg/kg, 10 mg/kg, 20 mg/kg, 50 mg/kg or 100 mg/kg. The dosage is likely to depend on such variables as the type and extent of progression of the disease or disorder, the overall health status of the particular patient, the relative biological efficacy of the compound selected, formulation of the excipient, and its route of administration.

(112) Effective doses can be extrapolated from dose-response curves derived from in vitro or animal model test systems. For example, in vitro analysis of cytokine production by peripheral blood mononuclear cells (PBMCs) can be a useful for assaying pro- and anti-inflammatory responses, e.g., secretion of IL-1β, IL-12, IL-4 or IL-10, IL-6, IL-23, and TNF-α.

Compositions can also be analyzed for effects in animal models, for example, IgA production, cytokine production by explants of Peyer's patches, and dendritic cell and T-cell responses.

(113) Wide variations in the needed dosage are to be expected in view of the variety of cellular targets and the differing efficiencies of various routes of administration. Variations in these dosage levels can be adjusted using standard empirical routines for optimization, as is well understood in the art. Administrations can be single or multiple (e.g., 2- or 3-, 4-, 6-, 8-, 10-, 20-, 50-, 100-, 150-, or more fold). Encapsulation of the compounds in a suitable delivery vehicle (e.g., polymeric microparticles or implantable devices) may increase the efficiency of delivery.

(114) The duration of treatment with any composition provided herein can be any length of time from as short as one day to as long as the life span of the host (e.g., many years). For example, a composition can be administered once a week (for, for example, 4 weeks to many months or years); once a month (for example, three to twelve months or for many years); or once a year for a period of 5 years, ten years, or longer. It is also noted that the frequency of treatment can be variable. For example, the present compositions can be administered once (or twice, three times, etc.) daily, weekly, monthly, or yearly.

(115) Any method known to those in the art can be used to determine if a particular response is induced. Clinical methods that can assess the degree of a particular disease state can be used to determine if a response is induced. For example, a subject can be monitored for symptomatic relief, e.g., relief from colic, diarrhea, constipation, nausea, vomiting, abdominal pain, cramping, heartburn, abdominal distention, flatulence, or incontinence, dermatitis, redness, pain, swelling. Alternatively or in addition, serum markers, imaging techniques, e.g., ultrasound, x-rays, and endoscopic methods can be used.

(116) The compositions may also be administered in conjunction with other therapeutic modalities. Other therapeutic modalities will vary according to the particular disorder, but can include, for example, anti-inflammatory agents, antibiotics and other dietary treatments, anti-diarrhea medications, anti-emetics, anti-cholinergic agents. Concurrent administration of two or more therapeutic agents does not require that the agents be administered at the same time or by the same route, as long as there is an overlap in the time period during which the agents are exerting their therapeutic effect. Simultaneous or sequential administration is contemplated, as is administration on different days or weeks.

(117) Articles of Manufacture

(118) The invention also features kits for administering the compositions. Accordingly, packaged products (e.g., sterile containers containing one or more of the compositions described herein and packaged for storage, shipment, or sale at concentrated or ready-to-use concentrations) and kits, are

also within the scope of the invention. A product can include a container (e.g., a vial, jar, bottle, bag, microplate or beads) containing one or more compositions of the invention. In addition, an article of manufacture further may include, for example, packaging materials, instructions for use, syringes, delivery devices, buffers or other control reagents.

(119) For example, the kit can include a measured amount of a composition comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effect or polypeptide went to a second bacterial effector polypeptide and a protein transduction domain. The compounds, agents, and/or reagents can be packaged in a suitable container. The kit can further comprise instructions for administering the compositions. For example, the kit can include: a fusion protein comprising a protein transduction domain and a set of paired peptides. The kit can also include a buffering agent, a preservative, and/or a protein stabilizing agent. Each component of the kit can be enclosed within an individual container and all of the various containers can be within a single package. The product may also include a legend (e.g., a printed label or insert or other medium describing the product's use (e.g., an audio- or videotape or computer readable medium)). The legend can be associated with the container (e.g., affixed to the container) and can describe the manner in which the reagents can be used. The reagents can be ready for use (e.g., present in appropriate units), and may include one or more additional adjuvants, carriers or other diluents.

(120) Also included as effectors or effector proteins of the compositions and methods described herein are functional equivalents of the proteins described above. By the term “functional equivalent” is meant any amino acid sequence or modification thereof that has the same targeting and immune suppressing function of the naturally occurring effector protein. In one embodiment, such functional equivalents can have modifications of one or more amino acids from the known sequences. In one embodiment, such functional equivalents can be a smaller fragment of the known sequences. In one embodiment, such functional equivalents can be a derivative of the naturally occurring sequences or be derived from other than human sources. In one embodiment, such functional equivalents can be altered by chemical modification or be altered by recombinant production to be associated with sequences with which the effector proteins are not associated in nature. Similarly, chemical or structural changes or fragments of the nucleic acid sequences that encode the effector proteins are also considered functional equivalents herein.

(121) As used herein, the term “construct” as described herein refers to a chemically synthesized or genetically engineered assemblage that comprises one or more PTD/CPP associated with one or more effector proteins and further optionally associated with one or more targeting moieties. The construct can be in the form of a polypeptide or a nucleic acid molecule encoding the polypeptide.

(122) As used herein the term(s) “cosmeceutically or pharmaceutically acceptable carrier, excipient or formulation” refer to the components of a composition that provide a vehicle for delivery. For example, where the cosmeceutical or pharmaceutical product is a topical composition, the carrier or formulation can contain typical components such as cremes, saline, vitamins, oils that are normally found in cosmetic or pharmaceutical preparations for skin. See, for example, U.S. Pat. No.

5,635,497 which discloses an oil-in-water, fatty cream composition for topical administration comprising from 60 to 80 percent by weight of fatty components, from 1.5 to 5 percent by weight of at least one non-ionic, hydrophilic surfactant having an HLB of at least 14, about 6% of fatty alcohols and esters, a therapeutically effective amount of at least one topically active therapeutic agent, and water, provided the topically active therapeutic agent is not dithranol or its derivatives. See, also International Patent Publication No. WO2014/076642. Pharmaceutically or cosmetically acceptable excipients suitable for the compositions described herein can be selected from plasticizers, disintegrants, glidants, coloring agents, lubricants, stabilizers, adsorbents, preservatives, delivery retarders and mixtures thereof. Such a composition may contain a transepidermal or transdermal carrier agent consisting of acidic electrolyzed water having a pH of 1.0 to 4.0 and comprise clusters of water having 5 to 10 molecules of water per cluster, and a polyacrylate. Other suitable formulations may include oils, emollients, lotions for topical and

transdermal applications along with buffered/aqueous and saline solutions. See, also, texts such as *Topical Drug Delivery Formulation* (eds. Osborne and Amann), 2000, publishers Taylor & Francis, *Drugs and the Pharmaceutical Sciences Series #42*.

(123) The term “CAGE” as used herein refers to deep eutectic synthetic solvent, a choline-based oil that has antimicrobial activity, which has been shown to penetrate deep into the dermis. CAGE is described by Zakrewsky, M. et al, *Adv. Healthcare Mater.* (March 2016), 5, 1282-1289, incorporated by reference herein. CAGE has been shown to be useful in transdermal protein delivery, wherein the protein is carried 15-20 cell layers into model skin preparations. See, e.g., Banerjee, A. et al, *Adv. Healthcare Mater.* 2017, 1601411 DOI: 10.1002/adhm.201601411, incorporated by reference herein.

(124) Still other pharmaceutical strategies to enhance dermal delivery of peptides or proteins including carrier peptides, signaling peptides neurotransmitter-inhibiting peptides and enzyme-inhibiting peptides, include chemical and physical penetration enhancers, such as listed in Table 2 of Badenhurst, T. et al, *Pharmaceutical Strategies for the Topical Dermal Delivery of Peptides/Proteins for Cosmetic and Therapeutic Applications*. *Austin Journal of Pharmacology and Therapeutics* (2014), 2(6):10. Also as discussed in this document are coupling with lipophilic moiety, such as lauric, palmitic and other acids, using CPP conjugates, formulation with microemulsions, encapsulation in liposomal vesicles and use of lipid particles, as well as combinations of these formulations.

(125) The term “targeting moiety” refers to constructs useful in fusion with the effector proteins and/or PTD/CPPs described herein to direct the fusion protein or a nucleic acid sequence encoding it to a specific cell or tissue type within the body. Alterations to the fusion constructs described by the addition of amino acid segments to naturally occurring effector or fusion effector sequence, which enables the protein to bind and to specifically target cells, tissues or other target or physiologic compartment in the human body create “targeted-effectors” and “targeted effector fusions”. Such targeting moieties include amino acid segments that enhance the efficacy of the preparation by its ability to be activated due to conditions in a specific compartment of the body could be cleaved, for instance by proteolytic cleavage, addition of post-translational modifications, or forming of PPIs with cell and tissue specific host proteins. Examples of tissue-specific targeting peptides for this use include those described in Jung, E. et al., *Identification of tissue-specific targeting peptide*. *J Comput Aided Mol Des* (October 2012) 26:1267-1275, incorporated by reference herein. Targeting moieties can also be antibodies, antibody fragments, aptamers, amino acid sequences, nucleic acid sequences that are complementary to or capable of binding a complementary sequence on a cell or tissue or chemical moieties that have a three-dimensional structure that can fit into a three-dimensional pocket on the targeted cell or tissue. For example, a targeting sequence can be a hormone, or fragment thereof, that targets or binds its naturally occurring cell surface receptor, or tissue specific markers, etc.

(126) The term “polypeptide,” when used in singular or plural form, generally refers to a polymer of amino acids joined together by peptide bonds and may include unmodified or naturally occurring amino acids or modified or unnatural amino acids. In certain embodiment, the term polypeptide refers to a construct formed by multiple shorter peptides joined directly, or indirectly via linkers, to form a single peptide. In one embodiment, as described herein, a polypeptide is formed by the fusion of a PTD or CPP and an effector protein. In another embodiment, as described herein, a polypeptide is formed by the fusion of a PTD attached to an effector protein, wherein the effector protein is further attached to another effector protein (with or without its own PTD). In still another embodiment, the polypeptide of this invention is formed by the covalent association of a first PTD fused to a first effector protein, a linker followed by an additional (e.g., second, third, fourth, etc) effector protein. The term “first” is used only to distinguish among the effectors. In yet another embodiment, each effector protein of the polypeptide is associated with its own PTD or CPP. In yet another embodiment, only one PTD/CPP is present. In another embodiment, the single PTD/CPP is

located at the N terminus of the polypeptide. In yet another embodiment, multiple PTD/CPP are present in the polypeptide, each located at the N terminus of its associated effector protein. In another embodiment, each PTD/CPP is separated from its effector protein by a linker. In another embodiment, each effector protein is separated from each additional effector protein by a linker. The first and additional effector constructs in a single polypeptide may occur in any order.

(127) In still another embodiment of a polypeptide as described herein, the PTD/CPP are fused directly to the effector protein and each effector protein is fused to each additional effector protein. In certain embodiments, the polypeptides contain two or more different effector proteins, each having its own target. In certain embodiments, the polypeptides contain two or more different effector proteins, each targeting related targets. In still further embodiments, each polypeptide is further associated with a targeting moiety to target the polypeptide to a specific tissue or cell type, e.g., skin, epidermis, dermis.

(128) By “homologous protein” is meant a protein having a percent sequence similarity or identity of greater than 80%, greater than 85%, greater than 90%, greater than 95%, greater than 97%, or greater than 99% and sharing the same function as the effector protein.

(129) As used herein, the term “polynucleotide,” when used in singular or plural form, generally refers to any polyribonucleotide or polydeoxribonucleotide, which may be unmodified RNA or DNA or modified RNA or DNA that encode any of the polypeptide constructs as described above. Thus, for instance, polynucleotides as defined herein include, without limitation, single- and double-stranded DNA, DNA including single- and double-stranded regions, single- and double-stranded RNA, and RNA including single- and double-stranded regions, hybrid molecules comprising DNA and RNA that may be single-stranded or, more typically, double-stranded or include single- and double-stranded regions. In addition, the term “polynucleotide” as used herein refers to triple-stranded regions comprising RNA or DNA or both RNA and DNA. The term “polynucleotide” specifically includes cDNAs. The term includes DNAs (including cDNAs) and RNAs that contain one or more modified bases. In general, the term “polynucleotide” embraces all chemically, enzymatically and/or metabolically modified forms of unmodified polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including simple and complex cells.

(130) By “nucleic acid molecule” as used herein is meant the nucleic acid sequence that encodes a construct or polypeptide as described above. The nucleic acid molecule can include other operative components, such as regulatory sequences directing expression of the construct or polypeptide in a cell in vivo or in vitro. The nucleic acid molecule can be in a vector.

(131) By “vector” is meant an entity that delivers the nucleic acid molecule to cells, for therapeutic or cosmetic purposes. As used herein, a vector may include any genetic element including, without limitation, naked DNA, a phage, transposon, cosmid, episome, plasmid, or a virus. Vectors are generated using the techniques and sequences provided herein, in conjunction with techniques known to those of skill in the art. Such techniques include conventional cloning techniques of cDNA such as those described in texts such as Sambrook et al, *Molecular Cloning: A Laboratory Manual*, 3rd edition, 2001 Cold Spring Harbor Press, Cold Spring Harbor, N.Y., and current editions thereof, use of overlapping oligonucleotide sequences of the adenovirus genomes, polymerase chain reaction, CRISPR, gene editing, and any suitable method which provides the desired nucleotide sequence.

(132) By the term “attachment” or “attach” as used herein to describe the interaction between the components of the constructs is meant covalent attachments or a variety of non-covalent types of attachment. Still another useful attachment mechanism involves via “affinity interactions”, i.e., one domain fused to an antibody fragment that recognizes an epitope on the second domain to be used instead of the two domains fused together. Other attachment chemistries useful in assembling the constructs described herein include, but are not limited to, thiol-maleimide, thiol-haloacetate, amine-NHS, amine-isothiocyanate, azide-alkyne (CuAAC), tetrazole-cyclooctene (iEDDA).

(133) The “linker” refers to any moiety used to attach or associate different elements of the polypeptide/polynucleotide sequence components of the constructs (i.e., the effector, the PTD, the targeting moiety) to each other. Thus in one embodiment, the linker is a covalent bond. In another embodiment, the linker is a non-covalent bond. In an embodiment of a polynucleotide described herein, the linker is composed of at least one to about 20 nucleic acids. Thus, in various embodiments, the linker is formed of a sequence of at least 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57 up to about 60 nucleic acids. In yet another embodiment of a polypeptide as described herein, the linker refers to at least one to about 20 amino acids. Thus, in various embodiments, the linker is formed of a sequence of at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or about 20 amino acids. In still other embodiments, the linker can be a larger compound or two or more compounds that associate covalently or non-covalently. In still other embodiment, the linker can be a combination of the linkers defined herein. The linkers used in the constructs of the compositions and methods are in one embodiment chemically or enzymatically cleavable, such as by redox, pH, and the like. The linkers used in the constructs of the compositions and methods are in one embodiment non-cleavable.

(134) The term “nucleic acid molecule” refers to a recombinant assembled nucleic acid sequence encoding a construct or polypeptide described herein. The nucleic acid molecule may be naked DNA or RNA. Alternatively the nucleic acid molecule may be associated operatively with regulatory sequences permitting expression of the construct or polypeptide in vivo or in vitro. The nucleic acid molecule may be a vector, plasmid vector, or be presented in a viral vector for delivery to the subject. Generation of such nucleic acid molecules with resort to the teachings of this specification can utilize known recombinant and genetic engineering techniques. See, e.g., “Inflammatory conditions” as used herein refer, in one aspect to inflammatory skin diseases, which are the most common problem in dermatology and cause pain, redness, swelling and the sensation of heat. Such inflammatory skin conditions include, without limitation, non-specific rashes accompanied by skin itching and redness, sunburn, dermatitis, eczema, rosacea, seborrheic dermatitis, psoriasis, infection, skin injury or wounds, autoimmunity or aging effects. Other inflammatory conditions (non-skin) include, without limitation, autoimmune conditions, asthma, chronic peptic ulcer, tuberculosis, rheumatoid arthritis, periodontitis, ulcerative colitis and Crohn's disease, sinusitis, active hepatitis, gut dysbiosis syndromes and any other disease caused by NFkB/JNK/p38 pathway activation.

(135) “Patient” or “subject” as used herein means a mammalian animal, including a human, a veterinary or farm animal, a domestic animal or pet, and animals normally used for clinical research. In one embodiment, the subject of these methods and compositions is a human.

(136) The terms “a” or “an” refers to one or more. For example, “an expression cassette” is understood to represent one or more such cassettes. As such, the terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

(137) As used herein, the term “about” means a variability of plus or minus 10% from the reference given, unless otherwise specified.

(138) The words “comprise”, “comprises”, and “comprising” are to be interpreted inclusively rather than exclusively, i.e., to include other unspecified components or process steps. The words “consist”, “consisting”, and its variants, are to be interpreted exclusively, rather than inclusively, i.e., to exclude components or steps not specifically recited.

(139) With resort to the definitions of the components above, in one aspect, the inventors provide a composition comprising a first construct comprising a selected immunomodulatory effector protein or its functional equivalent thereof that targets a first functional domain optionally linked covalently or non-covalently to a selected protein transduction domain (PTD) or penetrating peptide (CPP); an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP of (a) or to an additional PTD or CPP, or a combination of a first and one or more additional

constructs.

(140) In one embodiment, the first and additional constructs are further linked with a targeting moiety to direct activity of the composition to a specific cell or tissue, e.g., skin. In another embodiment, the constructs are admixed in a pharmaceutically or cosmeceutically acceptable carrier or excipient or formulation, such as a formulation suitable for topical administration to the skin.

(141) One embodiment of such a composition is a single polypeptide comprising a fusion of two or more effectors. Another embodiment is a single polypeptide comprising a first effector-PTD fused construct linked covalently or non-covalently to one or more of additional constructs. The polypeptide in a further embodiment contains one or more optional linker amino acid sequences interposed between each construct of the polypeptide. In one embodiment, the polypeptide contains fused to the first construct and/or additional construct, a targeting moiety to direct the polypeptide to a specific cell or tissue. In a further embodiment, the single polypeptide is in a pharmaceutically or cosmeceutically acceptable carrier or excipient or formulation.

(142) Whether the constructs are admixed in a composition or present in one or more polypeptides, the effector can be one or more of the effectors identified above, particularly in Tables 1 or 2. In one embodiment of the admixture or polypeptide, the first effector protein is NleE or a functional equivalent thereof. In yet other embodiment, the additional effector protein is one or more of NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In other embodiments, the admixture of constructs in the composition or present in the single polypeptide can be selected from the following embodiments. In one embodiment, the first effector is NleE and the additional effector is NleC. In another embodiment, the first effector is NleE and the additional effector is NleD. In another embodiment, the first effector is NleE and the additional effector is NleB. In another embodiment, the first effector is NleE and the additional effector is NleH. In another embodiment, the first effector is NleE and the additional effector is YopB. In another embodiment, the first effector is NleE and the additional effector is YopH. In another embodiment, the first effector is NleE and the additional effector is YopJ. In another embodiment, the first effector is NleE and the additional effector is YopP. In still another embodiment, the first effector is NleE and the additional effector is SspH1. In a further embodiment, the first effector is NleE and the additional effector is OspG. In another embodiment, the first effector is NleE and the additional effector is OspF. In another embodiment, the first effector is NleE and the additional effector is IpaH9.8. In still a further embodiment, the first effector is NleE and the additional effector is IpaH1.4. In another embodiment, the first effector is NleE and the additional effector is IpaH2.5. In another embodiment, the first effector is NleE and the additional effector is IpaH4.5. In another embodiment, the first effector is NleE and the additional effector is IpaH7.8. In yet a further embodiment, the first effector is NleE and the additional effector is SlrP. In still other embodiments of the constructs or polypeptides, three or more effectors can be delivered in a single mixture of constructs or single polypeptide. The first and additional effectors or effector constructs may occur in any order in the fusions or in the polypeptides described herein.

(143) In the above-listed constructs and polypeptides, a suitable PTD or CPP linked to the first construct and optionally to each additional construct or polypeptide is the naturally occurring PTD of the selected effector protein, e.g., the YopM PTD (SEQ ID NO: 2; amino acids 1-50 of YopM). See U.S. Pat. No. 8,840,901.

(144) The PTD/ CPP can also be a functional equivalent, e.g., a mutated or modified version of a naturally occurring PTD, such as the naturally occurring PTD sequence of SspH1, or of *Shigella* IpaH protein or any other of the effectors. The PTD useful in the constructs and polypeptides of the compositions can be a completely novel sequence which is developed to transport the construct or polypeptide or composition across cell membranes.

(145) In one specific embodiment, the first construct is the YopM PTD fused to NleE. In another

specific embodiment, the first construct is the SspH1 PTD fused to NleE. In still another specific embodiment, the first construct is the IpaH PTD fused to NleE. In still other embodiments, the additional construct comprises YopM, optionally associated with the YopM PTD. In still other embodiments of the single polypeptide, the polypeptide further contains at least one linker sequence as defined above interposed between the first and the one or more additional constructs. In still other embodiments of the constructs and single polypeptide, each construct or polypeptide can further contain a fused targeting moiety, such as a skin cell targeting peptide or other cell or tissue targeting peptide. Given the number of effectors, PTD/CPPs, linkers and targeting moieties identified herein and in the cited publications, any number of constructs or polypeptides may be prepared according to the teachings contained herein.

(146) In still other embodiments of the above-listed constructs and polypeptides, the admixture of protein constructs or constructs contained on the single polypeptide is such that the targeted functional domain of each effector protein in the construct is expressed in the same cell. In another embodiment, each construct in admixture or in the single polypeptide has a non-overlapping redundant role in inhibiting NFkB, JNK and p38 pathways when present in a mammalian cell.

(147) As described above, any of the compositions described herein, whether a mixture of constructs, a single polypeptide or nucleic acid molecules encoding them, can be prepared in a formulation comprises ingredients suitable for application to, and absorption through, the cells of the skin. One such formulation employs the above-described CAGE solvent. Other suitable formulations may include oils, emollients, lotions for topical and transdermal applications along with buffered/aqueous and saline solutions.

(148) In yet another aspect, the compositions described herein include nucleic acid molecules comprising a nucleic acid sequence encoding a first construct, an additional construct, or the single polypeptide of any one of the above-described embodiments. Such nucleic acid molecules can also comprise operatively associated regulatory sequences, such as promoters, enhancer, etc (see, e.g., Sambrook et al) necessary to express the construct or polypeptide in a suitable cell. In certain embodiments, the nucleic acid molecule is naked DNA or RNA. In certain embodiments, the molecule is part of a plasmid or contained in a recombinant vector or virus. Methods for generating such nucleic acid molecules are within the skill of the art given the teachings herein. The nucleic acid molecules may also be delivered in a cosmeceutically or pharmaceutically acceptable carrier or excipient or formulation. Such formulations are described in detail in available texts as described above.

(149) These constructs and/or polypeptides may be formulation into a wide variety of cosmeceutically or pharmaceutically acceptable carrier or excipient or formulation for many different uses and routes of administration. While topical administration is preferred for uses on skin inflammation, it is contemplated that other conventional routes of administration will be used for treating other inflammatory conditions. In some embodiments, routes of administration include transdermal (including patch formulation), intra-dermal injection (including infusion and subcutaneous injection). Other pharmaceutically acceptable routes of administration include, but are not limited to, systemic routes, such as intraperitoneal, intravenous, intranasal, intramuscular, intratracheal, subcutaneous, epidural, and oral routes and other parenteral routes of administration or intratumoral or intranodal administration. The composition may be administered by any convenient route, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and may be administered together with other biologically active agents. Administration can be systemic or local. In addition, it may be desirable to introduce the pharmaceutical compounds or compositions of the invention into the central nervous system by any suitable route, including intraventricular and intrathecal injection; intraventricular injection may be facilitated by an intraventricular catheter, for example, attached to a reservoir, such as an Ommaya reservoir. Pulmonary administration can also be employed, e.g., by use of an inhaler or nebulizer, and formulation with

an aerosolizing agent. Routes of administration may be combined, if desired.

(150) Other methods of delivery of the effectors include via an attenuated bacterial strain which expresses a functional Type-Three Secretion System (TTSS)-expressing microorganism and is engineered to contain nucleic acid sequences encoding at least one of the effector immunomodulatory proteins or fusions or polypeptide, wherein amino acid sequences also code for functional TTSS secretion signal sequences. These include attenuated bacteria engineered or induced to shed outer membrane vesicles (OMV) or other type of exosome-like, bacterial or cell-derived vesicle containing the proteins described herein.

(151) In still other embodiments, the compositions, constructs, fusions and polypeptides and nucleic acid molecules described herein may be further manipulated by encapsulation in liposomes, micro particles, microcapsules, or in recombinant cells capable of expressing the compound, receptor-mediated endocytosis construction of a nucleic acid as part of a retroviral or other vector, etc.

(152) Still other embodiments of compositions are provided herein. In certain embodiments, a composition of single effectors, or fusions of multiple effectors, or fusions of single effector-PTD, or fusions of multiple effectors and one or more PTD, or single/multiple effector(s)-PTD-targeting moiety fusions are prepared. The polypeptide linker regions and other non-native peptide sequences are engineered into fusion-effectors and can contain active segments which can lead to their enhanced ability to be purified in active form. In another embodiment, the effector fusions can be modified recombinantly or chemically, or by e.g., editing methodologies to provide altered subcellular localization, altered stability, or altered ability to participate in protein-protein interactions. In one embodiment, the effector fusions are modified to include amino acid segments normally used as tags for efficient purification of proteins from complex mixtures (e.g., 6HIS, GST, and Maltose BP).

(153) In still a further specific embodiment, each individual effector protein or multiple effector fusion has an amino acid sequence that is deliberately altered by designed site directed mutagenesis. In one embodiment, such mutagenesis alters the activity of the effector in terms of its potency as an anti-inflammatory effector. In another embodiment, the altered effector amino acid sequence alters the effector's specificity for a target, catalytic activity, antigenicity, stability, ability to bind substrate, and/or ability to work with other effectors in a preparation. Such mutagenic techniques can alter any other common property of an effector protein for the benefit of the efficacy of the resulting therapeutic preparation.

(154) In another aspect, the compositions containing the effector fusions can lead to the inhibition of at least one or more of an NF- κ B, JNK and p38 mediated signal transduction pathway protein in selected cells and tissues. Such inhibition provides an anti-inflammatory response, an anti-apoptotic effect or a pro-apoptotic effect in a target cell.

(155) Still other embodiments of compositions can be obtained by selection of the components as taught herein resulting in a pharmaceutical/cosmeceutical preparation, containing active and therapeutic levels of the effector fusions and constructs as described herein for use in preventing, treating, or ameliorating an immune-related disorder whose pathology stems from aberrant activation of the NF κ B, JNK or P38 pathways.

(156) In still other aspects, methods for designing the various constructs and polypeptides described herein can be made using techniques well-known in the art of recombinant genetic engineering and manipulation of the nucleic acid techniques. Suitable techniques are known to those of skill in the art, and as provided by many of the publications incorporated by reference herein. Selection of the appropriate techniques to create the various embodiments will depend upon the effectors, PTDs, linkers, targeting moieties and other components as set out above. Molecular biology strategies to alter the proteins to make them more stable, specific, catalytic and robust are known. Further such known techniques can be used to optimize the design, construction and delivery of effector fusions based upon knowledge of the specific pathways and enzymes used by

the cell in a selected inflammatory stimulus.

(157) For example, in one embodiment the selected effectors are subject to random, high throughput mutagenesis in the catalytic active sites or the sites of substrate binding. Selection of effectors of these mutagenized populations for variants with novel substrate specificities, rates of catalysis and their combination with other known or mutagenized effectors in the fusions, constructs and polypeptides described herein can result in compositions with novel signal transduction inhibition functions.

(158) Still other known and available methods can be employed for creating recombinant fusion effector genes/proteins and for their expression, purification, stabilization as single effectors or fusion effectors. Such methods include producing/expressing the effector protein or effector fusion proteins from cloned genes in any suitable cell based systems, including, without limitation, bacteria, yeast, insect, or mammalian cells. Methods for purification of the resulting fusion constructs resulting in highly active, stable proteins which retain the ability to target their native substrate also employ known techniques. Additional techniques for producing compositions of this invention include, without limitation, optimization of the effector preparation by incorporation into liposomes, PEGylation of the effector constructs or fusions or polypeptides. The effector constructs may be subjected to chemical derivatization to associate or conjugate the construct physically or in admixture with additional entities. Such entities include without limitation, lipids, liposomes, other drugs, and cargo molecules including nucleic acids, polypeptides, organic molecules, small organic molecules, metals, nano-particles, viruses, modified viruses, viral vectors, antibodies and/or plasmids as targets for conjugation.

(159) Yet further aspects provide a variety of methods of using the compositions, fusions, constructs and polypeptides described above in methods for treating cosmetic conditions, such as inflammation or irritation due to normal skin agent, to the treatment of other inflammatory conditions, such as cancer, gut dysbiosis syndromes, and other conditions identified herein caused by NFkB/JNK/p38 pathway activation. For example one therapeutic use is to ameliorate the inflammation in the tumor microenvironment (TME) in any malignant or non-malignant condition that involves activation of the NFkB/JNK/p38 pathway.

(160) The various compositions defined above may be employed therapeutically in the down regulation/normalization of the immune response elicited by the ectopic or pathologic activation of NFkB, JNK and P38 pathways. In one embodiment, such methods of treating or suppressing pathway activation can occur in subjects suffering from the consequences of normal, naturally occurring conditions such as aging associated inflammation. In yet another method, the compositions are administered for treatment of numerous conditions wherein the three pathways are aberrantly activated resulting in an inflammatory response. Such aberrant responses include without limitation, infection, wound healing, reactive dermatitis, auto-immune disease, and malignant or non-malignant proliferative disorder. These diseases, conditions and syndromes wherein the three pathways are aberrantly activated also include, but are not limited to diseases caused by autoimmunity of the patient, topical inflammation, chronic inflammation, gastroenteritis, chronic gastritis, inflammatory bowel diseases (IBD), colitis ulcerosa, psoriasis, allergic reactions, Crohns disease, dysbiosis syndromes, cancer (including glioblastoma) rheumatoid arthritis related bone diseases characterized by changes in bone resorption, reduction and relief of the signs and symptoms associated with treating inflammation, and/or suppression of the immune system.

(161) These methods involve administering via a suitable route of administration (as described above) an amount of the composition (i.e., the effector fusions, effector-PTD fusions, effector-PTD-Targeting fusions, and the other embodiments described above) in an amount sufficient to reduce activation of the desired pathway.

(162) Desirably, the methods further involve, in one aspect, administering a suitable dose or doses of the fusion construct(s) in a therapeutic regimen to the subject in need thereof. In one embodiment, such administration can occur once or more prior to, simultaneously with, or after any

conventional additional treatment for the conditions. In one embodiment, where the condition is an infection, the additional component is an antibiotic. In another embodiment, wherein the condition is a cancer, the additional component is radiation or chemotherapy. Still other known “additional components” may be selected by one of skill in the art or the attending physician depending on the condition being treated and the physical status of the subject.

(163) Methods for determining the timing of frequency of administration will include an assessment of subject's response to the first administration of the composition. The dose is generally the lowest dose of the composition that is effective to suppress activation of the NF κ B, JNK/p38 pathway. In still other embodiments, a suboptimal dose is delivered in a continuous infusion or a slow release formulation. The dosage required will depend primarily on factors such as the condition being treated, the age, weight and health of the patient, and may thus vary among patients. In one embodiment, where the composition comprises an admixture of peptide constructs or the single polypeptide, one such dose is about 1 to 25 μ M protein/polypeptide. In another embodiment, the dose is less than 10 μ M protein/polypeptide. In still another embodiment, the dose is between 1 μ M and 5 μ M protein/polypeptide. In another embodiment, the suboptimal dose is less than 1 μ M protein/polypeptide.

(164) When the composition is in the form of a nucleic acid or vector or nucleic acid molecule, it is administered in sufficient amounts to transduce the targeted cells and to provide sufficient levels of gene transfer and expression to reduce or inhibit activation of the NF κ B, JNK/p38 pathway and provide a therapeutic benefit without undue adverse or with medically acceptable physiological effects, which can be determined by those skilled in the medical arts. Dosages of these therapeutic compositions will depend primarily on factors such as the condition being treated, the age, weight and health of the patient, and may thus vary among patients. For example, a therapeutically effective adult dosage of a viral vector or siRNA nanoparticle is generally in the range of from about 100 μ L to about 100 mL of a carrier containing concentrations of from about 1×10^5 to about 1×10^{15} particles, about 1×10^{11} to 1×10^{13} particles, or about 1×10^9 to 1×10^{12} virus particles.

(165) In still another aspect, methods for use of the compositions described herein involve veterinary use for the treatment of inflammatory conditions in animals, e.g., for auto-immune diseases, reactive dermatitis, pruritis, alopecia and any other skin ailment for which the underlying pathogenesis involves alteration of NF κ B, JNK/p38 pathway. Selection of administration routes, dosages and therapeutic regimens may be selected by a veterinarian.

(166) In one aspect, a composition in a pharmaceutically acceptable carrier or excipient or formulation is provided. The composition includes: (a) a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain optionally linked covalently or non-covalently or by affinity to a selected protein transduction domain (PTD) or penetrating peptide (CPP); (b) an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP of (a) or to an additional PTD or CPP, or (c) a combination of constructs (a) and (b) in any order.

(167) In one embodiment of the composition, the first construct is linked covalently or non-covalently to one or more of the additional constructs in a single polypeptide. Additionally, the composition further includes an optional linker amino acid sequence interposed between each construct.

(168) In one embodiment, the functional equivalent includes a chemically or recombinantly modified amino acid sequence of the effector protein, or a fragment of the naturally-occurring effector amino acid sequence, or of the derivative of said chemically or recombinantly modified amino acid sequence of the effector protein that shares the functional activity of the effector protein.

(169) In one embodiment, the first effector protein is NleE or a functional equivalent thereof.

(170) In one embodiment, the additional effector protein is one or more of NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In one embodiment, the PTD or CPP is the naturally occurring PTD of the selected effector protein. In one embodiment, the PTD is a sequence of YopM, of SspH1, or of *Shigella* IpaH protein. In one embodiment, the PTD is amino acids 1-50 SEQ ID NO: 2.

(171) In one embodiment, the PTD or CPP is one or more of Poly-Arg, Tat and VP22, df Tat, a cyclic CPPs, IMT-P8, seven arginine (R7) and Streptolysin O (SLO)-mediated systems, elastin like polypeptide, CPP-adaptor system, 1,2-Benzisothiazolin-3-one (BIT) and Tat, activatable cell-penetrating peptides, LDP12, M918, BR2, POD, native protein independent of R11-CPP, Poly-arginine/Tat and Tat-PTD, Pep-1, CADY-2, R8, azo-R8, Penetratin, HR9 and IR9 peptides, or pVEC.

(172) In one embodiment, each targeted functional domain of each effector protein in the construct is expressed in the same cell. In one embodiment, each construct has a non-overlapping redundant role in inhibiting NFkB, JNK and p38 pathways when present in a mammalian cell. In one embodiment, the formulation comprises ingredients suitable for application to, and absorption through, the cells of the skin. In one embodiment, the formulation comprises CAGE solvent. In one embodiment, the first construct is the YopM PTD fused to NleE or the SspH1 PTD fused to NleE or the IpaH PTD fused to NleE. In one embodiment, the additional construct comprises YopM, optionally associated with the YopM PTD. In one embodiment, the composition further includes at least one linker between the first and additional constructs.

(173) In one embodiment, the first effector is NleE and the additional effector is NleC. In another embodiment, the first effector is NleE and the additional effector is NleD. In another embodiment, the first effector is NleE and the additional effector is NleB. In another embodiment, the first effector is NleE and the additional effector is NleH. In another embodiment, the first effector is NleE and the additional effector is YopB. In another embodiment, the first effector is NleE and the additional effector is YopH. In another embodiment, the first effector is NleE and the additional effector is YopJ. In another embodiment, the first effector is NleE and the additional effector is YopP. In another embodiment, the first effector is NleE and the additional effector is SspH1. In another embodiment, the first effector is NleE and the additional effector is OspG. In another embodiment, the first effector is NleE and the additional effector is OspF. In another embodiment, the first effector is NleE and the additional effector is IpaH9.8. In another embodiment, the first effector is NleE and the additional effector is IpaH1.4. In another embodiment, the first effector is NleE and the additional effector is IpaH2.5. In another embodiment, the first effector is NleE and the additional effector is IpaH4.5. In another embodiment, the first effector is NleE and the additional effector is IpaH7.8. In another embodiment, the first effector is NleE and the additional effector is SlrP. In any of the embodiments above, the constructs may be in any order.

(174) In one embodiment, the composition further includes a skin cell targeting peptide.

(175) In another aspect, a nucleic acid construct is provided. The nucleic acid sequence includes a nucleic acid sequence encoding the polypeptide of any of the compositions described above.

(176) In one embodiment, the nucleic acid construct further includes regulatory sequences necessary to express the polypeptide in a suitable cell.

(177) In one embodiment, the nucleic acid construct is a DNA, RNA, a plasmid or a recombinant vector or virus.

(178) In one embodiment, the nucleic acid construct is in a pharmaceutically acceptable carrier or excipient or formulation.

(179) In another aspect, a method is provided for treating, preventing or ameliorating an NFkB, JNK or p38 mediated inflammatory disorder in a subject, or a subject at risk from developing such disorder. The method includes administering to the subject any of the compositions or nucleic acid constructs described above.

(180) In one embodiment of the method, the composition is administered in an amount sufficient to

down-regulate the innate inflammatory response in a targeted cell or tissue of the subject mediated by the pathologic or non-pathologic activation of intracellular NFkB, JNK or P38 signal transduction pathway.

(181) In one embodiment, the method reduces symptoms, and inhibits progression, of damage to the subject's cells or tissue caused by inflammation.

(182) In one embodiment of the method, a composition containing multiple constructs targets multiple targeted functional domains that are expressed in the same cell and each construct plays multiple, non-overlapping redundant roles in inhibiting NFkB, JNK and p38 pathways.

(183) In one embodiment of the method, the presence of multiple constructs in the compositions produces a synergistic therapeutic effect.

EXAMPLES

Example 1: Preparation of Recombinant Constructs

(184) Sequences. The native YopM, NleE, OSPZ, and IpA7.8 nucleotide sequences were codon optimized for efficient bacterial expression of the encoded polypeptides. The optimized nucleotide sequences were synthesized de novo as long single stranded nucleotide chains. The overlapping and complementary single strand DNA sequences were annealed and the mixture subject to polymerize chain reaction (PCR) to generate full-length double-stranded (ds) DNA. The ds DNA was gel isolated and cloned into plasmid pQE60. The sequence of the inserted DNA was confirmed by nucleotide sequence analysis.

(185) TAT-NleE wild type and NleE Mutant R107A. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37° C. The Vector was then phosphatase treated for 1 hour then heat inactivated prior to ligation. All material was run through agarose gels and gene cleaned using a Qiagen gel extraction kit. Ligation reaction was performed at 16° C. overnight, and half of the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 for 1 hr at 37° C. to test for the presence of the insert. 675 bp Positive clones were then digested using EcoR5 (internal to the insert) and external Hind3 site for 1 hr at 37° C. to test the orientation of the insert. Positive orientation yielded a drop-out that was 175 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction purifications under native conditions. A schematic of the construct (based upon the backbone plasmid pQE60 available from Qiagen Inc, Catalogue number: #32903) is shown in FIG. 3.

(186) TAT-*Shigella* OSPZ. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37°C. The Vector was then phosphatase treated for 1 hr then the enzyme was heat inactivated prior to ligation. All material was electrophoresed through agarose gels and the DNA isolated from the gel using a Qiagen DNA gel extraction kit. Ligation reaction was performed at 16°C overnight, and half of the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 37°C for 1 hr to test for the presence of the 900 bp insert. Positive clones were then digested using EcoR5 (internal to the insert) and Hind3 located after the HIS-tag 37°C 1 hr to test the orientation of the insert. Positive orientation yielded a drop-out DNA fragment that was 200 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction of proteins and purification of protein under native conditions. A schematic of the construct is shown in FIG. 4.

(187) YopM PTD-NleE wild type and Mutant R107A. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37°C. The Vector was then phosphatase treated for 1 hr and heat inactivated prior to ligation. All material was run through agarose gels and gene cleaned using a Qiagen gel extraction kit. Ligation reaction was performed at 16°C overnight, and half of

the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 at 37°C for 1 hr to test for the presence of the insert. Positive clones were then digested using Bgl2 (internal to the insert) and Bgl2 at 37°C for 1 hr to test the orientation of the insert. Positive orientation yielded a drop-out that was 275 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction purifications under native conditions. A schematic of the construct is shown in FIG. 5.

(188) YopM PTD-NleE wild type and Mutant R107A, No Linker, PAPA Linker, and GSGS Linker. A QE60 vector containing a codon optimized YopM sequence was purchased from Epoch Life Science Inc. The plasmid was digested using Cla1 (internal to YopM) and Bgl2 (located outside of the cloning site). NleE was then digested using Cla1 and BamH1. Note that after ligation, the BamH1/Bgl2 site on the vector was lost. Because the vector was opened at a site that was internal to YopM, the ligation of NleE (along with its mutant R107A counterpart) lead to the fusion of the two inserts. This site is located downstream to the PTD, thus leaving it intact. Variations of this construct were made using either PAPA or GSGS linkers. These were added to the beginning of NleE during PCR by having the bridges be part of the forward primer sequence. Screening of these various constructs were done via PCR of mini-preps using NleE Cla1 FOR and TNHis Bam REV primers designed for NleE. Positive clones were then sent for sequencing to verify the clones prior to transformation into S9-cells for protein expression. A schematic of the construct is shown in FIG. 6.

(189) Schematics of the constructs used are shown in FIG. 1 and FIG. 2.

(190) Purification of recombinant polypeptides. The transformed SG13009 [pREP4] were plated on an agar plate containing both ampicillin and kanamycin. A single colony was used to inoculate a 200 mL culture, shaking at 200 RPM at 37° C. overnight. A volume of the overnight culture was added to 1 L of 2YT medium containing ampicillin and kanamycin and grown as described above in an incubator shaker at 200 RPM 37° C. to an OD 600 nm of 0.6-0.8. Then, 1 mL of 1M Isopropyl β-D-1-thiogalactopyranoside (IPTG) was added to the culture and growth was continued overnight with shaking at 200 RPM at 20° C. The following morning, cells were collected by centrifugation at 6,000 RPM at 4° C. The pelleted cells were resuspended in 40 mL Sonication Buffer and sonicated on ice for two cycles of 4-5 minutes each. The sonification buffer had a pH of 7.5 and 1 L of sonification buffer included one bottle of PBS, 300 mM NaCl, 10% Glycerol, 5 mM Imidazole. The remainder of the volume was filled with ddH.sub.2O. The buffer was then filtered and 1 ml 0.1M PMSF and 1 ml (2 mg/ml) each of Aprotinin, Leupeptine, and Pepstatin were added fresh. The sonicate was centrifuged 12,000 RPM at 4° C. for 30 minutes to separate lysed cells from soluble protein.

(191) The supernatant was collected and incubated with pre-equilibrated Ni-NTA beads (available from BioRad) for 1 hour at 4° C. by rotation. The binding capacity of polypeptides for Ni-NTA beads was about 5-10 mg/ml. Following the binding step, bound materials were centrifuged 4,000 RPM 4° C. for 10 minutes to pellet the Ni-NTA beads and separate from unwanted materials or “flow-through”. Beads were then washed twice in 40 mL wash buffer by rotation in 4° C. for 30 minutes per wash cycle. The wash buffer had a pH of 7.5 and 1 L of wash buffer included one bottle of PBS, 300 mM NaCl, 10% Glycerol, and 20 mM Imidazole. The remainder of the volume was filled with ddH.sub.2O. In addition, 1 ml 0.1 M PMSF was added fresh. The washed Ni-Beads were loaded on to a disposable gravity drip column and washed with residual Wash Buffer.

(192) Bound proteins were eluted with elution buffer, using 2× elution buffer per 1 mL of packed volume of beads, i.e., 2 ml elution buffer for 1 ml of Ni-NTA beads. The elution buffer had a pH of 7.5 and 100 ml of elution buffer contained one bottle of PBS, 300 mM NaCl, 10% Glycerol, and 500 mM Imidazole. The remainder of the volume was filled with ddH.sub.2O. The elution buffer was filtered before use. Eluate was collected in 2 mL fractions. The protein content of the fractions

was analyzed using Bradford reagent. Individual fractions were dialyzed overnight in dialysis buffer at 4° C. The dialysis buffer had a pH of 7.5 and 4 L of elution buffer included one bottle of PBS, 300 mM NaCl, and 10% Glycerol. The remainder of the volume was then filled with ddH₂O. The following were freshly added to the elution buffer: 1 ml 0.1 M PMSF and 1 ml 1 M DTT. In the morning, the dialysis buffer was discarded and replaced with fresh buffer and dialysis was continued for an additional 3 hours. After dialysis, the individual fractions were pooled and centrifuged at 12,000 RPM at 4° C. for 20 minutes to remove any precipitation. The soluble protein was concentrated in the final concentration was determined using Bradford reagent. (193) Purified polypeptides were analyzed by SDS-polyacrylamide gel electrophoresis (SDS-PAGE) analysis using known volumes of bovine serum albumin (BSA) as standards. Concentrated samples were aliquoted and stored at -80° C. Proteins were used directly after thawing on ice. An SDS-PAGE analysis of purified recombinant chimeric effector proteins is shown in FIG. 7. Each lane contained 2 µg of protein. Proteins were visualized by Coomassie blue staining.

Example 3: Cytokine Release

(194) The effect of the bacterial effector proteins on cytokine release was analyzed in a cell-based assay using the human monocyte cell line, THP-1. Cytokine release was measured in vitro using the LEGENDplex Multi-Analyte Flow Assay Kit (BioLegend, Cat no. 740118) according to the supplier's directions. For culturing THP-1 cells, RPMI with glutamine and 20% FBS was used. These cells were incubated at 37° C. in humidified air 5% CO₂ atmosphere. Finally, cell medium was changed every 2-3 days, and the cells were passaged at 95% confluence. To allow the cells to adapt to their environment, 3×10⁵ cells were added to a 24-well plate the night before. The next day, protein was added to these cells for a period of 3 hours. The protein was then washed out with PBS. After cells were stimulated overnight with LPS stimulation, a concentration of 1 µg/mL was used. The next day, cells were washed again and cytokine were measured. All 13 inflammatory cytokines were measured using the LEGENDplex Human Inflammation Plane (13-plex) assay from Biolegend.

(195) We analyzed the effect of effector protein constructs on the release of the pro-inflammatory cytokines IL-1β, TNF-α, IL-6, MCP-1, IL 23 and IL-8. The effector protein constructs are shown in the table below:

(196) TABLE-US-00006 Protein SEQ Construct transduction ID name domain Effector 1 Linker Effector 2 NO YopM Full- YopM YopM 3 length TAT-NLeE TAT EPEC NLeE — — 28 TAT Shigella TAT Shigella OSPZ 34 OSPZ YopM PTD- YopM EPEC NLeE — — 37 NLeE YopM PTD- YopM YopM (L-rich) — EPEC 10 NLeE (No NLeE linker) YopM PTD- YopM YopM (L-rich) GSGS EPEC 16 NLeE (GSGS NLeE linker) YopM PTD- YopM YopM (L-rich) PAPA EPEC 13 NLeE (PAPA NLeE linker) YopM PTD- YopM YopM (L-rich) — — 7 YopM (L-rich) IpaH 7.8 IpaH 7.8 IpaH 9.8 IpaH 9.8

(197) The results of these experiments are shown in FIGS. 8, 9, 10, 11 and 12. As shown in FIGS. 8, 9, 10, 11 and 12, the effector proteins reduced the levels of cytokines released into the medium relative to control samples that did not receive effector proteins. This effect was dose-dependent, with higher doses of effector proteins resulting in a steeper reduction in levels of released cytokines. These dose-dependent effects are shown in FIG. 8 with respect to IL-6. Similar dose-dependent effects were observed for IL-1β, TN F-α, IL-6, MCP-1, IL 23, and IL-8.

(198) The effector fusion proteins that included more than one effector protein domain produced a steeper reduction in the level of all released cytokines than did native effector proteins. As shown in FIG. 9, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NLeE; YopM PTD-YopM (L-rich)-GSGS-NLeE; YopM PTD-YopM (L-rich)-PAPA NLeE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of TNF α released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

(199) As shown in FIG. 10, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NLeE; YopM PTD-YopM (L-rich)-GSGS-NLeE; YopM PTD-YopM (L-rich)-PAPA NLeE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of IL-6 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

(200) As shown in FIG. 11, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NLeE; YopM PTD-YopM (L-rich)-GSGS-NLeE; YopM PTD-YopM (L-rich)-PAPA NLeE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of MCP-1 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

(201) As shown in FIG. 12, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NLeE; YopM PTD-YopM (L-rich)-GSGS-NLeE; YopM PTD-YopM (L-rich)-PAPA NLeE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of IL-23 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

(202) Similar results were observed for IL-8. Taken together, the results shown in FIGS. 8, 9, 10, 11, and 12 indicate that fusion proteins that target multiple inflammatory pathways are more effective at reducing an inflammatory response than are proteins that target a single inflammatory pathway.

Example 4: Caspase Activity

(203) The effect of YopM on caspase activity was measured using R&D System's Caspase-1/ICE Colorimetric Assay Kit (K111-100) according to the supplier's instructions. THP-1 cells were incubated with rYopM or a truncated YopM (YopMo) and then stimulated with LPS (lipopolysaccharide) and ATP to induce activation of caspase 1. More specifically, 2×10^6 THP-1 cells were seeded in 6-well cell culture plates in triplicate. The cells were incubated for 2 hours with rYopM (25 μ g/ml) or a truncated YopM (YopMo) and then LPS was added (1 μ g/mL) and the cells were incubated for an additional 4 hours. Finally, ATP (5 mM) was added and the cells were incubated for 45 minutes. To pellet the cells, centrifugation was carried out at $250 \times g$ for 10 min. The cell pellet (2×10^6 cells) was resuspended in 50 μ l lysis buffer (Caspase 1/ICE Colorimetric Assay Kit R & D Systems) and incubated on ice for 10 min. Cell debris were pelleted at $10,000 \times g$ for 1 min. The colorimetric assay was carried out in 96-well plates. For this purpose, 50 μ l cell lysate/well were placed and mixed with 50 μ l $2 \times$ reaction buffer (Caspase 1/ICE Colorimetric Assay Kit R & D Systems). After addition of 5 μ l of the substrate Ac-YVAD-pNA, incubation was carried out at 37° C. for 3 h. Subsequently, the absorbance at 405 nm was measured.

(204) The results of this experiment are shown in FIG. 13. As shown in FIG. 13, a dose-dependent reduction of caspase 1 activity was observed in THP-1 cells that had been treated with rYopM. A dose-dependent reduction of caspase 1 activity was also observed in THP-1 cells that had been treated with the truncated YopM (YopMo). These data showed that the truncated YopM (YopMo) that contained only the L-rich region of the YopM retained the caspase I reducing function of the full length YopM.

Examples 5: Effector Polypeptide Uptake

(205) Confocal microscopy. Cell uptake of effector polypeptides was analyzed using confocal microscopy. We analyzed single cell uptake of the following fusion protein constructs:

(206) TABLE-US-00007 Protein SEQ Construct transduction ID name domain Effector 1 Linker Effector 2 NO TAT-NleE TAT EPEC NLeE — — 28 YopM PTD- YopM YopM (L-rich) — EPEC NLeE 10 NleE (No linker) YopM PTD- YopM YopM (L-rich) GSGS EPEC NLeE 16 NleE (GSGS

linker) YopM PTD- YopM YopM (L-rich) PAPA EPEC NleE 13 NleE (PAPA linker)

(207) Fusion proteins were labeled with FITC using the Sigma-Aldrich FluoroTag™ FITC Conjugation Kit according to the supplier's instructions as follows. Protein and FITC were dissolved in carbonate-bicarbonate buffer. The FITC was slowly added to the protein with stirring, then covered with foil and stirred for two hours at room temperature. The conjugate was separated from free FITC on a G-25 column in the fractions were collected. Fractions containing conjugate were pooled and the F/P ratio of conjugate was determined spectrophotometrically. The labeled protein was stabilized with 1% bovine serum albumin and 0.1% sodium azide and stored at 0-5° C.

(208) HaCat cells were cultured using DMEM High glucose medium with 10% FBS. These cells were incubated at 37° C. in humidified air 5% CO₂ atmosphere. Finally, cell medium was changed every 2-3 days, and the cells were passaged at 95% confluence. Cells were grown to appropriate confluency. Next, appropriate coated coverslips were added to well plates and the cells were seeded in the well plates containing the coverslips. For uptake analysis, polypeptides were added to the cells to a final concentration of 50 ug/mL and incubated for two hours. The protein with the cells were incubated and the protein was aspirated after the incubation period was complete. The cells were fixed with 4% formaldehyde and washed once again. The membrane was permeabilized with 0.2% triton-x-100 and washed again. The cells were blocked for 30 minutes and incubated in primary antibody (either Cy5 or Actin) overnight. The antibody was then aspirated, washed, and incubated in appropriate secondary antibody for 30 minutes. The cells were then washed and DAPI solution was applied for 5 minutes. The cells were then washed and mounted using Prolong Gold Anti-Fade Reagent. The mounted cells were allowed to cure overnight and then imaged. The DAPI was imaged using a wavelength of 490 nm. The Cy5 and Actin were imaged using 594 nm wavelength.

(209) Detection of protein from fixed cells was performed using Nikon 80i upright fluorescent microscope. A filter cube specific for the relative green range (~488 nm) was used to detect FITC labelled protein within cells.

(210) The results of this experiment are shown in FIGS. **14**, **15**, **16**, and **17**. As shown in FIG. **14**, incubation of cells with QE12-TAT-NleE resulted in nuclear fluorescence. As shown in FIGS. **15** and **16**, both QE60-YopM PTD-NleE (GSGS linker) and QE60-YopM PTD-NleE (PAPA linker), respectively, produced punctate cytoplasmic fluorescence that was not observed in untreated control cells (FIG. **17**). Taken together, these data show that the fusion polypeptides were taken up by individual cells.

(211) Inverted microscopy. Uptake of QE12-TAT-NleE was analyzed by inverted microscopy as follows. A segment of skin was isolated from a shaved section of a mouse, and the tissue was preserved in a saline solution to maintain the cell viability within the tissue. The tissue was stained with 50 ug of FITC labeled TAT-NleE that was applied to the top of the sample and allowed to diffuse over a 2 hour period. Afterwards, the tissue was fixed with paraformaldehyde and stained with DAPI. The skin segment was then sliced in 10 uM increments, and the top slice was observed in an inverted microscope was used with varying objectives. As shown in FIG. **18**, the TAT-NleE fusion polypeptide entered intact mouse skin and penetrated through multiple cell layers.

(212) Two-photon microscopy. Penetration of TAT-NleE- into mouse skin was analyzed by two-photon microscopy. A 1 cm×1 cm segment of mouse skin was stained with. 50 µg of FITC-labeled TAT-NleE for two hours and counter stained with DAPI. the sample was analyzed by two-photon microscopy. More specifically, mouse hair was removed post-euthanasia and a 1 cm×1 cm segment of skin was removed and submerged in 1×PBS pH 7.4 to maintain tissue viability. Afterwards, the PBS was removed, and 50 ug FITC-TAT-NleE was applied to the top of the skin and allowed to incubate at room temperature for two hours. Afterwards, the tissue was washed three times for five minutes each with 1×PBS pH 7.4 to remove all waste. From there the tissue was permeabilized with Triton-x and fixed with 4% PFA. DAPI was then added to solution of PBS at a concentration of 1:1000 and incubated in the tissue for 3-5 minutes. Then the washing period was repeated (three

times for five minutes each in 1 ml of PBS). The tissue was then mounted on a dish submerged in PBS ready for 2-photon imaging.

(213) A 3 dimensional image is shown in FIG. 19 and a 10 μ M slice of the stained tissue is shown in FIG. 20. As shown in FIGS. 19 and 20, the TAT-NleE penetrated the skin into the epidermis, but not the dermis.

Example 6: Methylase Activity

(214) Methylase activity of recombinant proteins that included NleE effectors was analyzed using the MTase glo assay from Promega. We assayed the methylase activity of wild-type NleE, mutant NleE R107A and fusion proteins YopM PTD-NleE (GSGS linker), YopM PTD-NleE (PAPA linker). We also analyzed the methylase activity of NLeE from *Shigella*.

(215) The in vitro methylase assay was developed using as a base, the MTase glo assay from Promega Inc. The primary assay buffer for enzyme activity was (25 mM Tris, pH 8, 50 mM NaCl, 1 mM EDTA, 3 mM MgCL₂, 0.1% BSA, 0.005% Tween20, 5 mM DTT). To determine the effect of various vehicles and buffers on NLeE a 40 μ L aliquot of vehicle was spiked with 1 μ L of 20 mg/mL NLeE (0.5 mg/mL final). After 30 mins, 5 μ L of spiked vehicle was diluted into 5000 μ L of primary assay buffer—(termed ‘SPIKED’ in graphs). To determine the effect of diluted vehicle on NLeE activity, 5 μ L of vehicle diluted to 5000 μ L of assay buffer was used. 1000 μ L was then removed and used for the no enzyme controls. 1 μ L of 0.5 mg/mL NLEE was then added to 1000 μ L of diluted vehicle and the assay commenced. The methyltransferase assay itself was performed in 4×1:2 serial dilutions of each sample generated above in assay buffer and 2.5 μ L added to 384-well assay plate. Reactions were initiated at various time points by the addition of 2.5 μ L of 2× MTaseGlo, 1 μ M GST-TAB2 protein substrate and 20 μ M SAM in assay buffer. After the time course, reactions were terminated by the addition of 5 μ L Methyltransferase Detection Reagent and the multiwell plates read for quantitative O.D. using a standard plate reader.

(216) The results of this analysis are shown in FIGS. 21, 22 and 23. As shown in FIG. 21, both fusion constructs retained the methylase functionality of native NLeE. As shown in FIG. 22, the presence of an R107A mutation in NLeE abolished methylase activity. As shown in FIG. 23, the *Shigella* NleE showed methylase activity at levels comparable to that seen for the EPEC NleE

Example 7: Effect of Formulations on NleE Methylation Activity

(217) We analyzed the effective three different commercial formulations/delivery agents on NleE methylation activity in vitro. These were designated: Formula A, Formula B, and Formula C

(218) The experimental setup was exactly as described above except that the buffer controls contained an equivalent amount of Formula A, Formula B or Formula C as controls. The results of this experiment are summarized in FIG. 24. As shown in FIG. 24a, NLeE methylase activity was retained in the presence of all three formulations. As shown in FIG. 24b, NLeE was stable in the presence of each of the formulations, although the degree of stability varied.

Example 8: How Specific is NleE for Inhibiting NFkB?

(219) We cloned expressed and tested 64 other human C4 ZF containing proteins, among them the sequences identified in the table below. But few of them were modified by NleE. Thus, the specificity of NleE is extremely high.

(220) TABLE-US-00008 CAN15_HUMAN/7-26 WsCvr..CtflNpagqrqCsiC CAN15_HUMAN/48-67 WpCar..CtfrNflgkeaCevC CAN15_HUMAN/147-166 WaCpr..CtlhNtpvassCsvC CAN15_HUMAN/344-363 WsCak..CtlnNptvaprCsaC CAN15_HUMAN/416-435 WaCpa..CtlnNalrakhCaaC EWS_HUMAN/522-543 WqCpnpgCgnqNfawrteCnq Oncogene RNA Binder C FUS_HUMAN/426-447 WkCpnptCenmNfswrneCnq Oncogene Annealing C HOIL1_HUMAN/197-216 WqCpg..CtfinKpstrpgCemC Binds Ub Chains MDM2_HUMAN/303-322 WkCts..CnemNpplpshCnrC P53 regulation MDM4_HUMAN/304-323 WqCte..CkkfNspskryCfrC P53 regulation NEIL3_HUMAN/321-340 WtCvv..CtlinKpsskaCdaC Endonuclease8 Repair NPL4_HUMAN/584-603 WaCqh..CtfnNqpptghCem Ub Chaperone Erad C NRP1_YEAST/359-378 WnCps..CgfsNfqrtaCfrC Rna Binder No Fcn NRP1_YEAST/585-604

WkCst..CtyhNfaknvvClrC NU153_HUMAN/662-681 WqCdt..ClIqNkvtdnCiaC NUP153 Nuc Pore NU153_HUMAN/726-745 WdCdt..ClvqNkpeaikCvaC NU153_HUMAN/797-816 WeCsv..CcvsNnaednkCvs C NU153_HUMAN/855-874 WdCel..ClvqNkadstkClaC RBM10_HUMAN/217-236 WlCnk..CgvqNfkrrekCfkC RNA Binding RBM5_HUMAN/185-204 WlCnk..CclnNfrkrIkCfrC RNA Binding RBP2_HUMAN/1356-1375 WhCns..CslkNastakkCvsC RANBP2 Huge Nuc Pore RBP2_HUMAN/1419-1438 WdCsi..ClvrNeptvsrCiaC RBP2_HUMAN/1483-1502 WdCsa..ClvqNegsstkCaaC RBP2_HUMAN/1547-1566 WdCss..ClvrNeanatrCvaC RBP2_HUMAN/1610-1629 WdCsv.. ClvrNeasatkCiaC RBP2_HUMAN/1669-1688 WdCsv..ClvrNeasatkCiaC RBP2_HUMAN/1728-1747 WdCsv..ClvrNeasatkCiaC RBP2_HUMAN/1785-1804 WdCsv..CcvqNessslkCvaC RBP56_HUMAN/358-379 WvCpnpsCgnmNfarmnsCnq C RNF31_HUMAN/304-323 WhCaa..CamlNepwavlCva C RNF31_HUMAN/354-373 WaCqs..CtfeNeaaavlCsiC RNF31_HUMAN/413-432 WyCih..CtfcNsspgwvCvmC RYBP_HUMAN/25-44 WdCsv..CtfrNsaeafkCsiC SHRPN_HUMAN/352-371 WsCps..CtfiNapdrpgCemC TAB2_HUMAN/668-687 WnCta..CtflNhpalirCeqC NFKB pathway TAB3_HUMAN/687-706 WnCds..CtflNhpalnrCeqC NFKB pathway TX13A_HUMAN/380-399 WdCpw..CnavNfsrrdtCfdC

(221) A cysteine methylation assay was performed and shows that NleE does not utilize zinc fingers of known highly related zinc fingers. NleE targets NFKB signaling at a key step in TAK1 kinase activation. The CYS-4 Zinc Finger of TAB2 was required to bind free K63 Ub chains to activate TAK1 kinase, which thereby triggers NFKB signaling. The TAB2/3 Cys 4 Zinc-Finger binds K63 linked Ub chains: The metal chelated structure of the ZF required binding. NleE targets NFKB signaling by methylating a single Cysteine in the C4 ZF TAB2 of TAB2 abolishing K63 Ub sensing. Cys 673 in TAB2 is the target of NleE. This highly specific activity of NleE is repurposed in the compositions described herein to shut down the NFKB pathway and reduce inflammation.

Example 9: Penetration of Recombinant NleE into the Skin in an Ex Vivo Pig Skin Model

(222) A key claim and potential stumbling block to use of NleE as a skin anti-inflammatory compound is to show that this globular folded protein of MW 38 kDa is able to penetrate the dermis and epidermis in order to access cells in which the NFKB system is aberrantly activated. A recombinant construct was prepared by fusing the amino acid sequence of NleE to an N-terminal histidine, and covalently labelling that peptide with FITC dye, resulting in the peptide His-NleE-FITC. --This construct was mixed with the eutectic, choline-based solvent CAGE, as described above at a concentration above at a concentration of 1-10 mg/ml. The amount of peptide in CAGE or PBS was 1 mg/ml. The final CAGE concentration was 90% v/v. The mixture was applied to live pig skin being perfused in a Franz diffusion chamber at 37 degrees C. for 24 or 48 hours. The Franz diffusion chamber is described by Technical Brief 2009 vol. 10, Particle Sciences-Drug Development Services (Bethlehem, PA). After these time periods, skin was harvested, fixed and sections followed by microscopy to visualize the FITC dye. The results (FIG. 25) show that CAGE-solubilized NleE FITC was able to penetrate the stratum corneum, epidermis and dermis and that virtually every cellular compartment of the skin. Up to 20-30 cell layers were contacted.-

(223) In comparison to published reports by Banerjee et. al 2017, particularly FIG. 1 using CAGE to transport the globular proteins ovalbumin and bovine serum albumin, the NleE penetrates as well if not better than these test proteins. NleE solubilized in PBS and applied to the pig skin showed no transduction past the stratum corneum.

(224) In summary, this pre-clinical data provides support that recombinant NleE protein is stable in the eutectic solvent CAGE and can be delivered to multi layers of the skin locally in a topical manner. It also provides support for using NleE protein and the combinations described herein that contain an intrinsic PTD domain which renders the protein capable of penetrating cells and tissues and turning off inflammatory cytokines. The ability of these combinations of proteins to penetrate, in an apparently active form and inhibit production of a multitude of inflammatory cytokines, is reflected in its therapeutic effect on psoriatic lesions in a mouse skin model. These compositions

delivered in a cream topically, in therapeutic levels are not anticipated to appreciably enter the regional lymphatics.

(225) These observations strongly support the use of recombinant bacterial effector proteins containing PTDs (Table 1) are useful as a platform for inhibiting various accessible inflammatory disease processes in the body.

(226) Thus, methods and compositions for treating inflammation related to activation of the NFkB, JNK and p38 pathways, e.g., inflammatory conditions of skin, include a composition comprising a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain; and at least one additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain. The first construct is optionally linked to a selected protein transduction domain (PTD) or penetrating peptide (CPP) or to a targeting moiety. The additional construct is optionally linked to the same PTD or CPP of the first construct or to an additional PTD or CPP. In one embodiment, a combination of these constructs is provided. In another embodiment the first construct is fused to one or more additional constructs in a single polypeptide. The composition further comprise an optional linker amino acid sequence interposed between each construct and an optional targeting moiety.

(227) Each and every patent, patent application, and publication, including websites cited throughout specification are incorporated herein by reference. Similarly, the SEQ ID NOs which are reference herein and which appear in the appended Sequence Listing are incorporated by reference. While the invention has been described with reference to particular embodiments, it will be appreciated that modifications can be made without departing from the spirit of the invention. Such modifications are intended to fall within the scope of the appended claims.

Claims

1. A composition comprising a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide linked to a second bacterial effector polypeptide, wherein the protein transduction domain is a YopM protein transduction domain, an SspH1 protein transduction domain, or an 1paH protein transduction domain wherein the protein transduction domain and the set of paired peptides comprise a fusion protein and the amino acid sequence of the fusion protein is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.
2. The composition of claim 1, wherein the first bacterial effector polypeptide and second bacterial effector polypeptide are different.
3. The composition of claim 1, wherein a first bacterial effector polypeptide and the second bacterial effector polypeptide are immunomodulatory.
4. The composition of claim 1, wherein the first bacterial effector polypeptide and the second bacterial effector polypeptide recognize a different molecular target or modulate a different inflammatory pathway.
5. The composition of claim 4, wherein the inflammatory pathway is the NFkB pathway, the JNK pathway, the p38 pathway or the STING pathway.
6. The composition of claim 1, wherein a linker is positioned between the first bacterial effector polypeptide and the second bacterial effector polypeptide.
7. The composition of claim 1, wherein the protein transduction domain is a YopM protein transduction domain.
8. The composition of claim 7, wherein the protein transduction domain comprises SEQ ID NO. 5.
9. The composition of claim 1, wherein (a) the first bacterial effector polypeptide is a YopM polypeptide or an NLeE polypeptide; or (b) the second bacterial effector polypeptide is a YopM polypeptide or an NLeE polypeptide.

10. The composition of claim 1, wherein the first bacterial effector polypeptide is a YopM polypeptide and the second bacterial effector polypeptide is an NLeE polypeptide.

11. A fusion protein comprising a set of paired peptides, wherein the set of paired peptides comprises a first bacterial effector polypeptide linked to a second bacterial effector polypeptide wherein the amino acid sequence of the fusion protein is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

12. A composition comprising a protein transduction domain polypeptide linked to a first bacterial effector polypeptide and at least one additional bacterial effector polypeptides wherein the first bacterial effector polypeptide is a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the at least one additional effector polypeptide is a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79.
